GRUNDFOS **INSTRUCTIONS**

Oxiperm® Pro

OCD-162

Installation and operating instructions

BE > THINK > INNOVATE >

Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products Oxiperm® Pro, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

— Machinery Directive (2006/42/EC).
  Standards used: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Low Voltage Directive (2006/95/EC).
  Standard used: EN 61010-1: 2001 (second edition).
— EMC Directive (2004/108/EC).
  Standards used: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions.

BG: EC декларация за съответствие

Ние, фирма Grundfos, заявяваме с пълна отговорност, че продуктите Oxiperm® Pro, за които се отнася настоящата декларация, отговарят на следните указания на Съвета за уеднаквяване на правните разпоредби на държавите членки на ЕС:

— Директива за машините (2006/42/EC).
  Приложени стандарти: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Директива за нисковолтови системи (2006/95/EC).
  Приложен стандарт: EN 61010-1: 2001 (второ издание).
— Директива за електромагнитна съвместимост (2004/108/EC).
  Приложени стандарти: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Тази ЕС декларация за съответствие е валидна само когато е публикувана като част от инструкциите за монтаж и експлоатация на Grundfos.

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne Oxiperm® Pro som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

— Maskindirektivet (2006/42/EF).
  Anvendte standarder: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Lavspændingsdirektivet (2006/95/EF).
  Anvendt standard: EN 61010-1: 2001 (anden udgave).
— EMC-direktivet (2004/108/EF).
  Anvendte standarder: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Denne EF-overensstemmelseserklæring er kun gyldig når den publiceres som en del af Grundfos-monterings- og driftsinstruktionen.

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte Oxiperm® Pro, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

— Maschinenrichtlinie (2006/42/EG).
  Normen, die verwendet wurden: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Niederspannungsrichtlinie (2006/95/EG).
  Norm, die verwendet wurde: EN 61010-1: 2001 (zweite Ausgabe).
— EMV-Richtlinie (2004/108/EG).
  Normen, die verwendet wurden: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Diese EG-Konformitätserklärung gilt nur, wenn sie in Verbindung mit der Grundfos Montage- und Betriebsanleitung veröffentlicht wird.

ES: Declaración CE de conformidad

Nosotros, Grundfos, declaramos bajo nuestra entera responsabilidad que los productos Oxiperm® Pro, a los cuales se refiere esta declaración, están conformes con las Directivas del Consejo en la aproximación de las leyes de las Estados Miembros del EM:

— Directiva de Maquinaria (2006/42/CE).
  Normas aplicadas: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Directiva de Baja Tensión (2006/95/CE).
  Norma aplicada: EN 61010-1: 2001 (segunda edición).
— Directiva EMC (2004/108/CE).
  Normas aplicadas: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Esta declaración CE de conformidad sólo es válida cuando se publique como parte de las instrucciones de instalación y funcionamiento de Grundfos.

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits DDA, DDC et DDE, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

— Directive Machines (2006/42/CE).
  Normes utilisées : EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Directive Basse Tension (2006/95/CE).
  Norme utilisée : EN 61010-1: 2001 (deuxième édition).
— Directive Compatibilité Electromagnétique CEM (2004/108/CE).
  Normes utilisées : EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Cette déclaration de conformité CE est uniquement valide lors de sa publication dans la notice d'installation et de fonctionnement Grundfos.

IT: Dichiarazione di conformità CE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti Oxiperm® Pro, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

— Direttiva Macchine (2006/42/CE).
  Norme applicate: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Direttiva Bassa Tensione (2006/95/CE).
  Norma applicata: EN 61010-1: 2001 (seconda edizione).
— Direttiva EMC (2004/108/CE).
  Norme applicate: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Questa dichiarazione di conformità CE è valida solo quando pubblicata come parte delle istruzioni di installazione e funzionamento Grundfos.

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten Oxiperm® Pro waarop deze verklaring betrekking heeft, in overeenstemming zijn met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG Lidstaten betreffende:

— Machine Richtlijn (2006/42/EC).
  Gebruikte normen: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Laagspannings Richtlijn (2006/95/EC).
  Gebruikte norm: EN 61010-1: 2001 (tweede editie).
— EMC Richtlijn (2004/108/EC).
  Gebruikte normen: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Deze EC overeenkomstigheidsverklaring is alleen geldig wanneer deze gepubliceerd is als onderdeel van de Grundfos installatie- en bedieningsinstructies.

PL: Deklaracja zgodności WE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby Oxiperm® Pro, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

— Dyrektywa Maszynowa (2006/42/WE).
Zastosowane normy: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Dyrektywa Niskonapięciowa (LVD) (2006/95/WE).
Zastosowana norma: EN 61010-1: 2001 (drugie wydanie).
— Dyrektywa EMC (2004/108/WE).
Zastosowane normy: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Deklaracja zgodności WE jest ważna tylko i wyłącznie wtedy kiedy jest opublikowana przez firmę Grundfos i umieszczona w instrukcji montażu i eksploatacji.

PT: Declaração de conformidade CE

A Grundfos declara sob sua única responsabilidade que os produtos Oxiperm® Pro, aos quais diz respeito esta declaração, estão em conformidade com as seguintes Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE:

— Directiva Máquinas (2006/42/CE).
Normas utilizadas: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Directiva Baixa Tensão (2006/95/CE).
Norma utilizada: EN 61010-1: 2001 (segunda edição).
— Directiva EMC (compatibilidade electromagnética) (2004/108/CE).
Normas utilizadas: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Esta declaração de conformidade CE é apenas válida quando publicada como parte das instruções de instalação e funcionamento Grundfos.

RU: Декларация о соответствии ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия Oxiperm® Pro, к которым относится настоящая декларация, соответствуют следующим Директивам Совета Евросоюза об унификации законодательных предписаний стран-членов ЕС:

— Механические устройства (2006/42/EC).
Применявшиеся стандарты: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Низковольтное оборудование (2006/95/EC).
Применявшийся стандарт: EN 61010-1: 2001 (второе издание).
— Электромагнитная совместимость (2004/108/EC).
Применявшиеся стандарты: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Данная декларация о соответствии ЕС имеет силу только в случае публикации в составе инструкции по монтажу и эксплуатации на продукцию производства компании Grundfos.

SI: ES izjava o skladnosti

V Grundfosu s polno odgovornostjo izjavljamo, da so naši izdelki Oxiperm® Pro, na katere se ta izjava nanaša, v skladu z naslednjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic ES:

— Direktiva o strojih (2006/42/ES).
Uporabljeni normi: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Direktiva o nizki napetosti (2006/95/ES).
Uporabljena norma: EN 61010-1: 2001 (druga izdaja).
— Direktiva o elektromagnetni združljivosti (EMC) (2004/108/ES).
Uporabljeni normi: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

ES izjava o skladnosti velja samo kadar je izdana kot del Grundfos instalacije in navodil delovanja.

SE: EG-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna Oxiperm® Pro, som omfattas av denna försäkran, är i överensstämmelse med rådets direktiv om inbördes närmande till EU-medlemsstaternas lagstiftning, avseende:

— Maskindirektivet (2006/42/EG).
Tillämpade standarder: EN 809: 1998, EN ISO 12100-1+A1: 2009, EN ISO 12100-2+A1: 2009.
— Lågspänningsdirektivet (2006/95/EG).
Tillämpad standard: EN 61010-1: 2001 (andra upplagan).
— EMC-direktivet (2004/108/EG).
Tillämpade standarder: EN 61326-1: 2006, EN 61000-3-2: 2006+A1: 2009+A2: 2009, EN 61000-3-3: 2008.

Denna EG-försäkran om överensstämmelse är endast giltig när den publiceras som en del av Grundfos monterings- och driftsinstruktion.

Pfinztal, 20 July 2012

ppa. U. Stemick

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Person authorised to compile technical file and empowered to sign the EC declaration of conformity.

Original installation and operating instructions.

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Warning

Prior to operation, read these installation and operating instructions.

1. General safety instructions

1.1 Purpose of these operating instructions

These operating instructions inform about operating and monitoring the system. For information on installation, maintenance, servicing and dismantling, please see the separate Service Instructions.

1.2 Symbols used in this document

The safety instructions are identified by the following symbols:

Warning

If these safety instructions are not observed, it may result in personal injury.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

Information about possible residual risks can be found:

- on warning signs located at the installation site
- at the beginning of each section in this manual
- immediately before steps associated with a residual risk.

1.3 Operators

Operators are persons who are responsible for operating and monitoring the system at the installation location. The system may only be operated by trained and qualified personnel.

The system is electronically controlled. Operators operate the system via a display with control elements.

Obligations of the operator

- Be trained by qualified personnel from Grundfos in the operation of the system.
- Observe the recognised regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals (Germany GUV-V D5).
- Keep secret the operator code for the operating software.

1.4 Obligations of the operating company

Owners of the building and/or operators of the Oxiperm Pro disinfection system are obliged to:

- Consider this manual to be part of the product and make sure that it is kept clearly accessible in the immediate vicinity of the system for the entire service life of the system.
- Meet the installation requirements specified by the manufacturer, see section *5.1*.
- Make sure that water lines and fittings are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Instruct operators in the operation of the system.
- Make sure that the labels supplied by the manufacturer are attached and clearly visible in the installation location.
  For illustration, see section *9. Photos*.
- Provide the operator code for the operating software only to operators who have received appropriate technical training.
- Make sure that the regulations for the prevention of accidents are observed in the installation location (Germany GUV-V D5).
- Provide all operators and service personnel with protective clothing in accordance with the regulations for the prevention of accidents (Germany GUV-V D5): face mask, gloves, protective apron, breathing mask.
- If the system is supplied without a dosing pump, provide an external dosing pump prior to installation. Only authorised service personnel trained by Grundfos is allowed to connect it to the Oxiperm Pro.

1.5 Authorised service personnel

The system may only be maintained and serviced by authorised service personnel trained by Grundfos.

1.6 Correct usage

Oxiperm Pro OCD-162 is a system for the discontinuous preparation of a chlorine dioxide solution out of hydrochloric acid (9 %) and sodium chlorite (7.5 %), and the continuous dosing of this solution for water disinfection.

1.7 Inappropriate usage

Applications other than those listed in section *1.6 Correct usage* are not in accordance with the intended use and are not permitted. The manufacturer, Grundfos, accepts no liability for any damage resulting from incorrect use.

A chlorine dioxide solution with an uncritical concentration of 2 g/l is generated in the reactor. Hence the Oxiperm Pro OCD-162 operates far out of the range of critical concentrations.

Danger of explosion in case of overdosage: at a concentration of more than 30 g/l the chlorine dioxide solution can explode.

Gaseous chlorine dioxide is a chemically unstable compound. At concentrations over 300 g/m^3, it decomposes into chlorine and oxygen explosively without external impact.

Warning

Unauthorised structural modifications to the system may result in serious damage to equipment and personal injury.

It is forbidden to dismantle, modify, change the structure of, bridge, remove, bypass or disable components, including safety equipment.

1.8 Safety and monitoring equipment

The system is fitted with the following safety and monitoring equipment:

- cover on the system frame
- two collecting trays for the two chemical containers (accessories)
- safety/multi-function valve at the dosing pump
- solenoid valve at the dilution water inlet
- volume compensation bag and activated-carbon filter for ClO_2 gas that escapes from the reaction tank
- alarm functions in the controller.

1.9 Chemicals

1.9.1 Chlorine dioxide concentration

In the reaction tank of the Oxiperm Pro OCD-162, diluted sodium chlorite and diluted hydrochloric acid are mixed to create a chlorine dioxide solution with a concentration of approximately 2 gram per litre of water. According to the disinfection requirement, the Oxiperm Pro OCD-162 doses the diluted chlorine dioxide solution into the main line to be disinfected. The chlorine dioxide concentration in drinking water must not exceed a maximum of 0.4 milligram per litre of water according to the German drinking water ordinance.

Warning

Risk of explosion when using chemicals in too high concentrations.

Only use sodium chlorite in a diluted concentration of 7.5 % by weight in accordance with EN 938.

Only use hydrochloric acid in a diluted concentration of 9.0 % by weight in accordance with EN 939.

Observe the safety data sheets from the chemicals supplier.

Warning

Risk of explosion when mixing up chemical containers or suction lances. Possible consequences are personal injury and serious damage to equipment.

Do not mix up chemical containers or suction lances.

Observe the labels on chemical containers, suction lances and pumps:
red = HCl, blue = $NaClO_2$.

Warning

Risk of burns when skin and clothing come into contact with sodium chlorite and hydrochloric acid.

Immediately wash affected skin and clothing with water.

Warning

Risk of irritation to eyes, respiratory system and skin, if chlorine dioxide is inhaled.

When changing the chemical containers, wear protective clothing in accordance with the regulations for the prevention of accidents (Germany GUV-V D5).

Warning

The temperature of the chlorine dioxide solution stored in an external batch tank may not exceed 40 °C.

Risk of outgassing at more than 40 °C.

Note

We recommend the installation of a gas warning device.

1.9.2 Storing chemicals

- Chemicals must be stored in the appropriately marked original plastic containers (20 to 33 litres).
- Do not store chemicals near grease, flammable substances, oils, oxidising substances, acids or salts.
- Empty and full containers must be kept closed, especially in areas where national regulations for the prevention of accidents apply to storage (Germany GUV-V D5).

1.9.3 Procedure in case of an emergency

The general safety regulations and regulations for the procedure in case of an emergency, as specified in EN 12671 apply.
Actions in case of an emergency:

- Ventilate the installation location immediately.
- Wear protective clothing (safety goggles, gloves, respirator and/or self-contained breathing apparatus, protective apron).
- Implement initial help measures
  - In case of contact with the eyes, rinse immediately with plenty of water for at least 15 minutes. Consult a doctor.
  - In case of contact with the skin, wash immediately with plenty of water. Remove all contaminated clothing.
  - In case gas is inhaled, move the casualty to a source of fresh air. Avoid taking deep breaths.
    Consult a doctor (look out for a racing pulse, as vasodilating treatment may be required).
- Spillages
  - In case of contact with clothing, remove the clothing immediately and wash with plenty of water.
  - Chemical spillages in buildings must be washed away with water.
  - Chlorine dioxide spillages can be doused with sodium thiosulphate, and washed away with water.
- Escaped gas
  - Escaped gas can be rinsed away with water from a sprinkling system.
- Firefighting
  - Aqueous solutions of chlorine dioxide are not directly flammable. Extinguish the surrounding fire with water, preferably using a fire sprinkler system to dilute the ambient gas. Inform the fire brigade of the installed production capacity and any harmful starting substances that are being stored (precursor substances) so that precautions can be taken regarding possible risks.

2. Product description

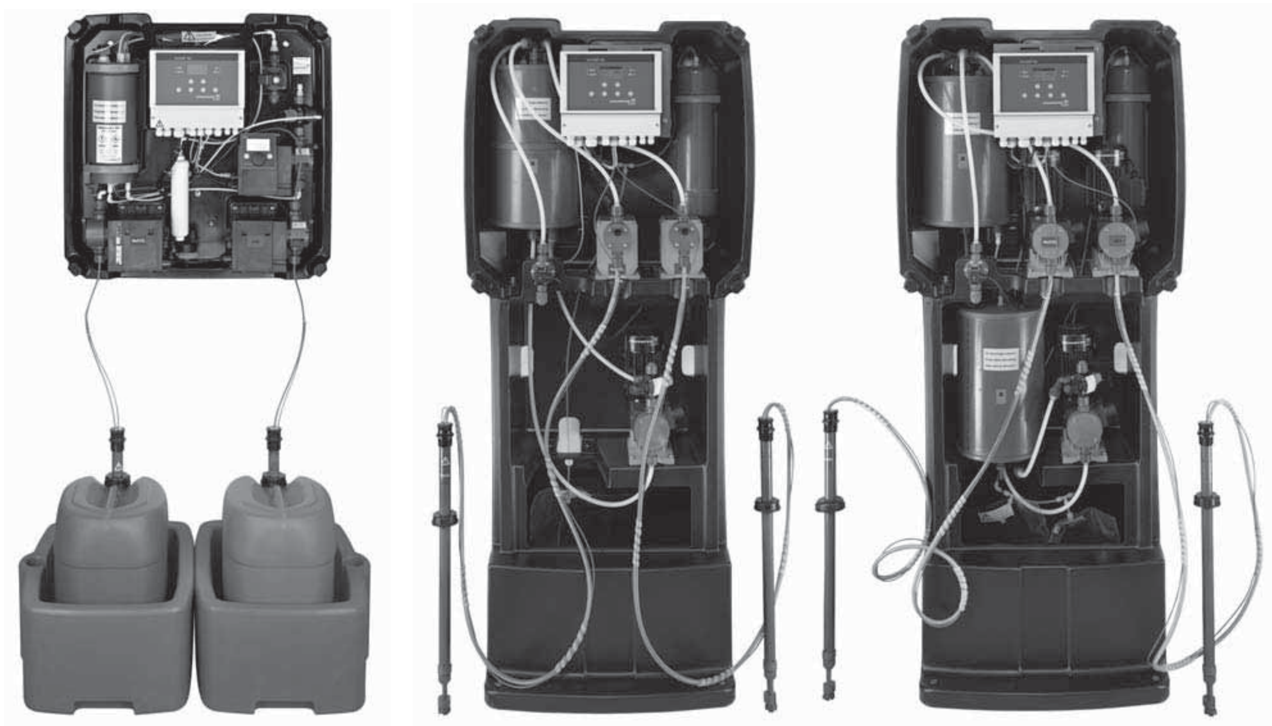

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Fig. 1 Oxiperm Pro OCD-162 without cover and peripheral devices

Oxiperm Pro OCD-162 disinfection systems produce and dose chlorine dioxide for the disinfection of drinking water, process water, cooling water and wastewater.

Oxiperm Pro OCD-162 disinfection systems consist of a plastic system frame, on which the internal components are mounted. They are prepared for wall- or floor-mounting, and covered by a plastic cover.

The chemicals are supplied from two original chemical containers, which are located in two collecting trays directly under the system (OCD-162-5, -10) or in a separate tray for each container next to the system (OCD-162-30, -60). A suction lance is inserted in each container and is permanently connected to the corresponding chemical dosing pump. The suction lance cables send low-level and empty signals to the controller.

Oxiperm Pro OCD-162 is connected to two water lines:

- The drinking water line for supplying dilution water and flushing water.
- The main water line to be disinfected, into which the final chlorine dioxide solution is dosed.

2.1 Application examples

Oxiperm Pro OCD-162 disinfection systems can be used for different types of applications:

2.1.1 Disinfection of drinking water lines

- The flow rate of the water in drinking water lines can fluctuate greatly (peak times when water is used for bathing and cooking).
- The type and level of contamination in the water (disturbance variables) are not known or are very varied.
- Examples: drinking water lines in:
  – hotels, multi-storey buildings
  – schools, hospitals, nursing homes
  – showers in sports facilities
  – food and beverage plants
  – waterworks.

2.1.2 Disinfection of industrial systems

- The water quantity in industrial systems is relatively constant.
- The type and level of contamination in the water (disturbance variables) are measured and hardly ever change.
- Examples:
  – bottle cleaning plants in breweries
  – industrial process water or wastewater systems
  – cooling water systems.

2.1.3 Shock disinfection (with external batch tank)

- Applications requiring large quantities of disinfectant in a short time
- Example: cleaning of whirlpool baths.

2.2 Functional principle

2.2.1 Production of chlorine dioxide

Chlorine dioxide is prepared in the reaction tank as follows:

Water, hydrochloric acid and sodium chlorite are added until a specific level is reached. During the reaction time, a diluted chlorine dioxide solution is produced. The reaction tank is filled with water. With a concentration of approximately 2 gram of chlorine dioxide per litre of water, the final solution flows through a pipe (overflow) located in the middle of the reaction tank into the batch tank below.

From the batch tank, the dosing pump doses the final chlorine dioxide solution to the injection unit, where it is injected into the main water line to be disinfected.

There are two different operation modes for the production of chlorine dioxide.

"Internal batch tank" mode: Chlorine dioxide solution is generated in the internal batch tank, and dosed into the line system, until the batch tank is empty. There are two ways of refilling the batch tank:

  – First method "1-20": You determine how many times the batch tank is refilled by choosing a number in the range of 1 to 20.
  – Second method "0 = continuous": The batch tank is refilled continuously.

- "External batch tank" mode: Chlorine dioxide solution is generated in the internal batch tank, and dosed into an external batch tank for storage. After emptying the external batch tank, chlorine dioxide production is restarted in a continuous process.

2.2.2 Flow-rate-proportional dosing

Suitable for drinking water applications:

1. The controller is set to "proportional controller".
2. A contact water meter or flow meter measures the water flow rate in the main water line, and continuously sends measured values to the controller.
3. The proportional controller calculates the required chlorine dioxide dosing volume in proportion to the water flow rate in the main line.
4. The proportional controller sends the corresponding output signals to the dosing pump.
5. The dosing pump doses the required quantity of chlorine dioxide solution from the batch tank into the main water line.
6. An optional measuring cell monitors the chlorine dioxide concentration in the main line.

2.2.3 Setpoint-controlled dosing

Suitable for industrial water applications:

1. The controller is set to "setpoint controller". A setpoint for the desired chlorine dioxide concentration in the main line is specified.
2. A measuring cell monitors the chlorine dioxide concentration in the main line, and sends actual values to the controller.
3. The setpoint controller compares the incoming actual values with the setpoint. Based on the deviation, it calculates the quantity of chlorine dioxide solution (actuating variable) required to achieve the desired concentration.
4. The setpoint controller sends output signals to the dosing pump.
5. The pump doses the corresponding quantity of chlorine dioxide solution from the batch tank into the main water line.

A combined controller is also available for applications with setpoint controller and flow meter, see separate Service Instructions.

2.3 Components of a standard system

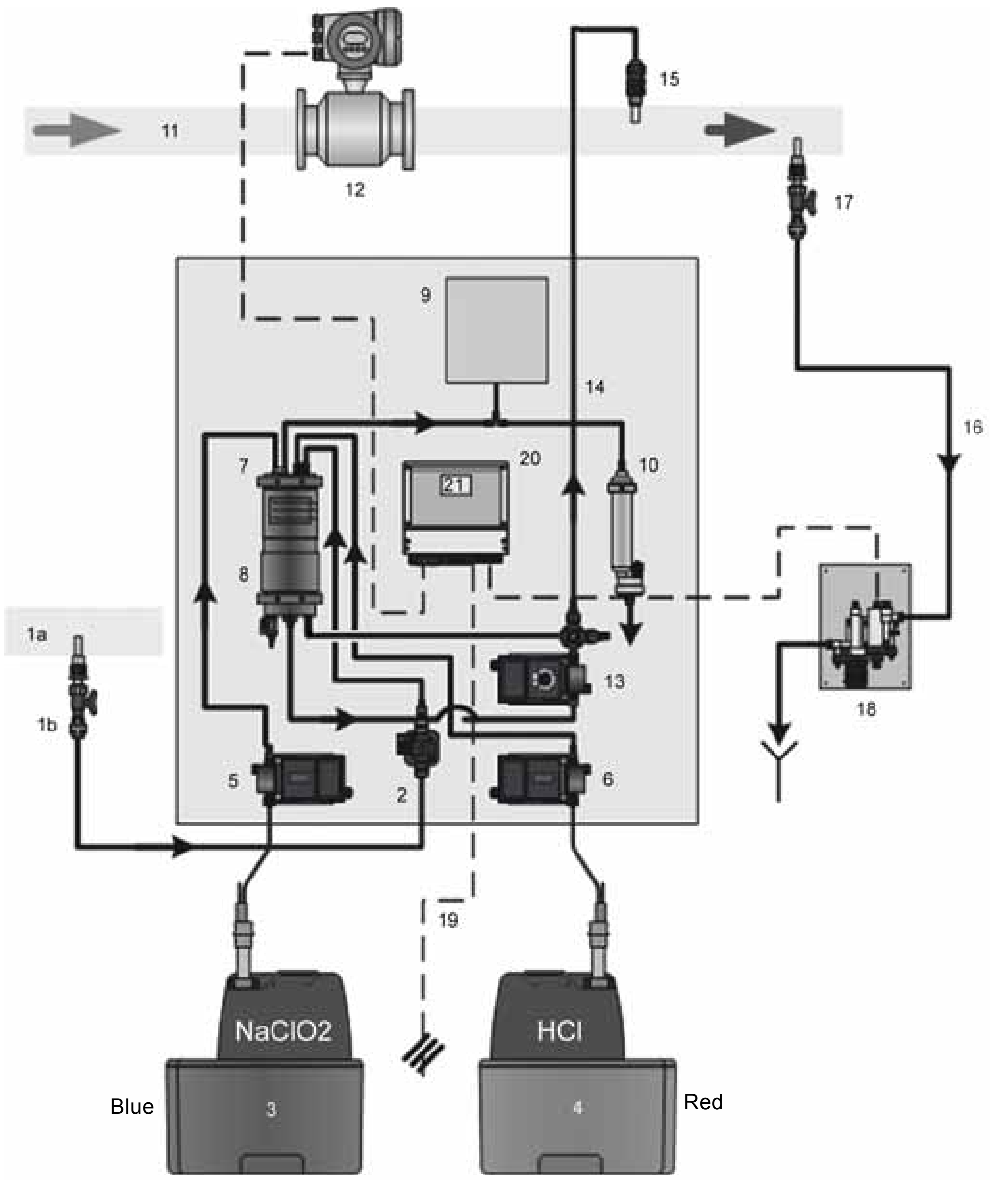

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Fig. 2 Components of a standard system (Oxiperm Pro OCD-162-5, -10)

2.3.1 External components

Pos.	Component
1a	Water line for dilution water and flushing water
1b	Dilution water extraction device with isolating valve
3	Container for sodium chlorite ($NaClO_2$, diluted concentration of 7.5 % by weight) with suction lance and collecting tray
4	Container for hydrochloric acid (HCl, diluted concentration of 9 % by weight) with suction lance and collecting tray
11	Main water line to be disinfected
12	Flow meter (or contact water meter)
14	Dosing line
15	Injection unit for dosing chlorine dioxide (ClO_2) into the main water line
16	Hose for sample water
17	Sample-water extraction device
18	Measuring cell for checking the chlorine dioxide concentration in the main water line (optional)
19	Power supply connection

See the photos in section *9. Photos*.

2.3.2 Internal components

Pos.	Component
2	Solenoid valve for dilution water and flushing water
5	Dosing pump for sodium chlorite ($NaClO_2$)
6	Dosing pump for hydrochloric acid (HCl)
7	Reaction tank with float switch
8	Batch tank with float switch and drain cock for chlorine dioxide (ClO_2)
9	Volume compensation bag for chlorine dioxide gas
10	Activated carbon filter for chlorine dioxide gas
13	Dosing pump with multi-function valve for chlorine dioxide
20	Electronic controller with measured-value sensor for check measurements
21	Display of the controller

2.4 Peripheral devices and accessories

2.4.1 Accessories for the dilution water line

- Isolating valve
- Dilution water extraction device (if necessary with double nipple and connection piece for hose).
- Hose with connection to solenoid valve.

Isolating valve and extraction device are not required, if the bypass mixing module with dilution water connection has been selected.

2.4.2 Accessories for the main water line

- Contact water meter or flow meter (in case of a new water line: water flow meter providing signals, or ultrasonic flow meter).
- Tapping sleeve for the injection unit.
- Protective pipe for the dosing line, installed from the dosing pump to the injection unit.
- DIT photometer: measures the chlorine dioxide concentration after dosing.
- Sample-water filter (in case of insufficient water quality).

2.4.3 Measuring cell

- Measuring cell
- Tapping sleeve for sample-water extraction at the main line.
- Hose from the sample-water extraction device to the measuring cell.
- Hose from the measuring cell to the sample-water drain.

2.4.4 Extension modules

The standard system can be extended with:

- Measuring cell for cold and hot water (main water up to 50 °C, pressure 4 bar).
- Measuring module for cold and hot water (main water up to 70 °C, pressure 8 bar).
- The use of a bypass mixing module is recommended in order to optimise mixing and reduce the risk of corrosion and in case of fluctuations in the main water flow

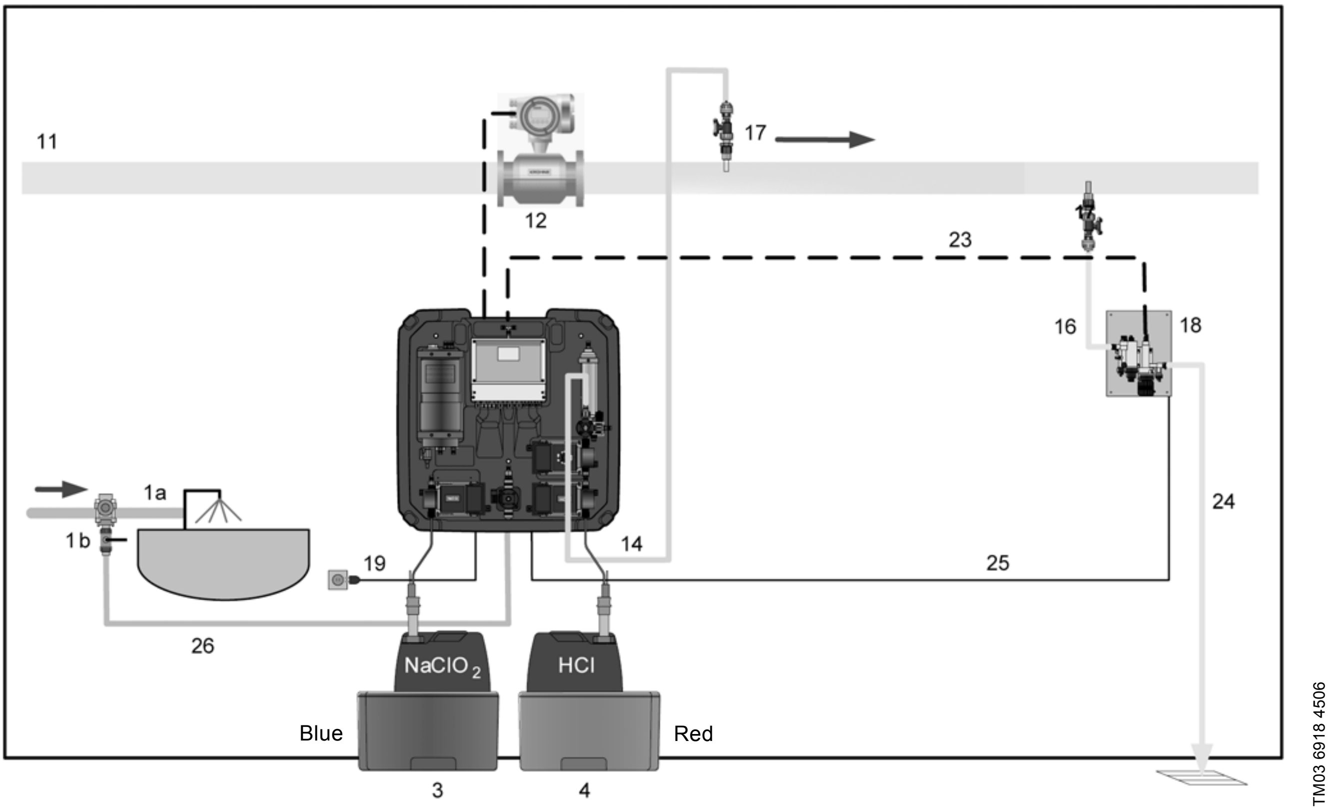

Fig. 3 OCD-162-5, -10 system with measuring cell and without extension module

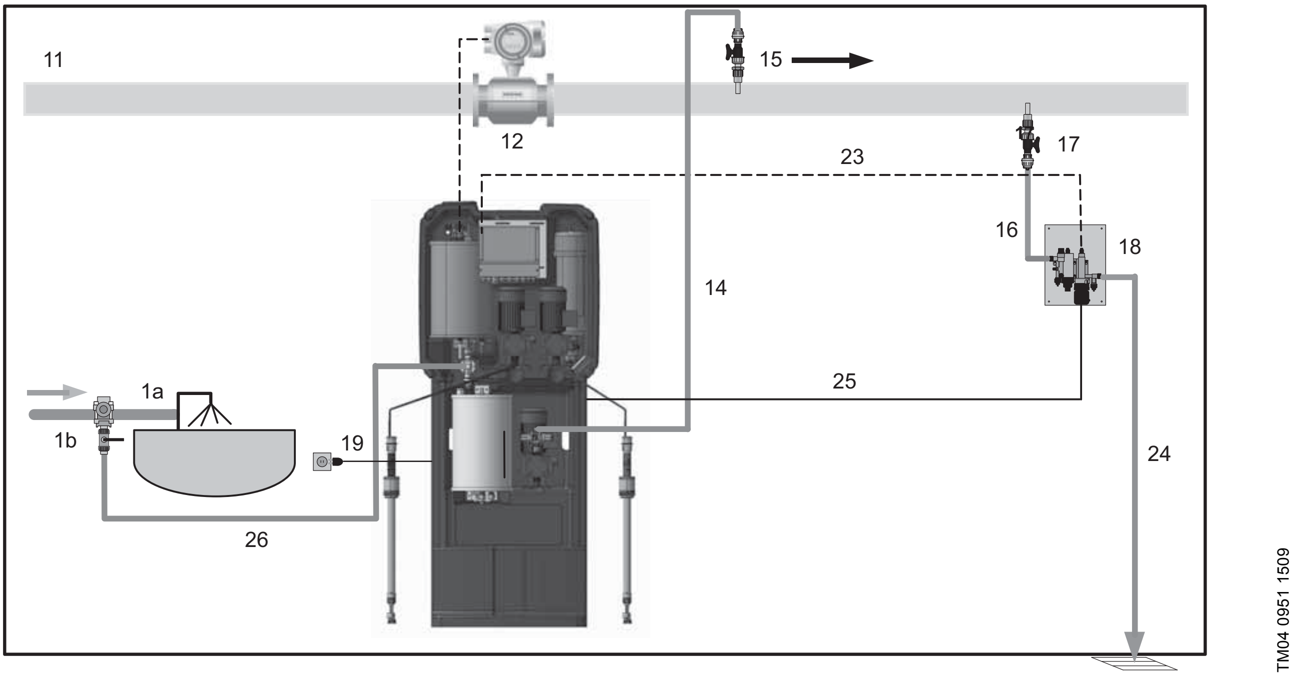

Fig. 4 Complete system with measuring cell (e.g. OCD-162-30, -60)

Pos.	Component
1a	Water line for dilution water and flushing water
1b	Dilution water extraction device with isolating valve
3	Container for sodium chlorite ($NaClO_2$, diluted concentration of 7.5 % by weight) with suction lance and collecting tray
4	Container for hydrochloric acid (HCl, diluted concentration of 9 % by weight) with suction lance and collecting tray
11	Main water line to be disinfected
12	Flow meter (or contact water meter)
14	Dosing line
15	Injection unit for dosing chlorine dioxide (ClO_2) into the main water line
16	Hose for sample water
17	Sample-water extraction device
18	Measuring cell
19	Power supply connection/main switch
23	Connection cable for measuring cell
24	Sample-water drain
25	Connection cable for cleaning motor
26	Hose for dilution water

2.5 Hydraulic connections

2.5.1 Oxiperm Pro OCD-162-5, -10

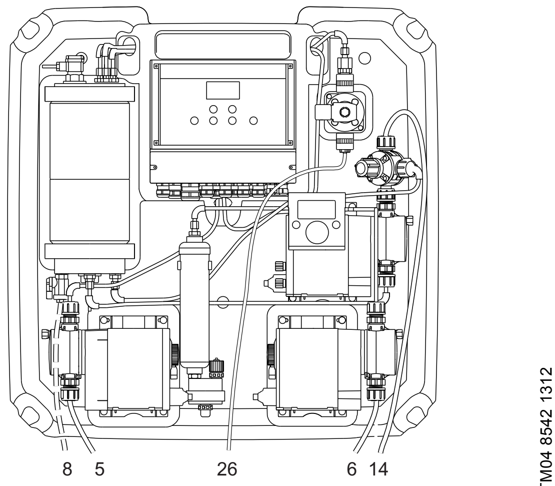

Fig. 5 Hydraulic connections OCD-162-05, -10

Pos.	Description
5, 6	Hoses for both suction lances on the suction side of the $NaClO_2$ and HCl dosing pumps
8	Hose at the drain cock of the batch tank (only installed for flushing and venting)
14	Dosing line from the chlorine dioxide dosing pump to the injection unit at the main line or to the injection unit at the mixing module or to the external batch tank
26	Dilution water hose at the solenoid valve

2.5.2 Oxiperm Pro OCD-162-30, -60

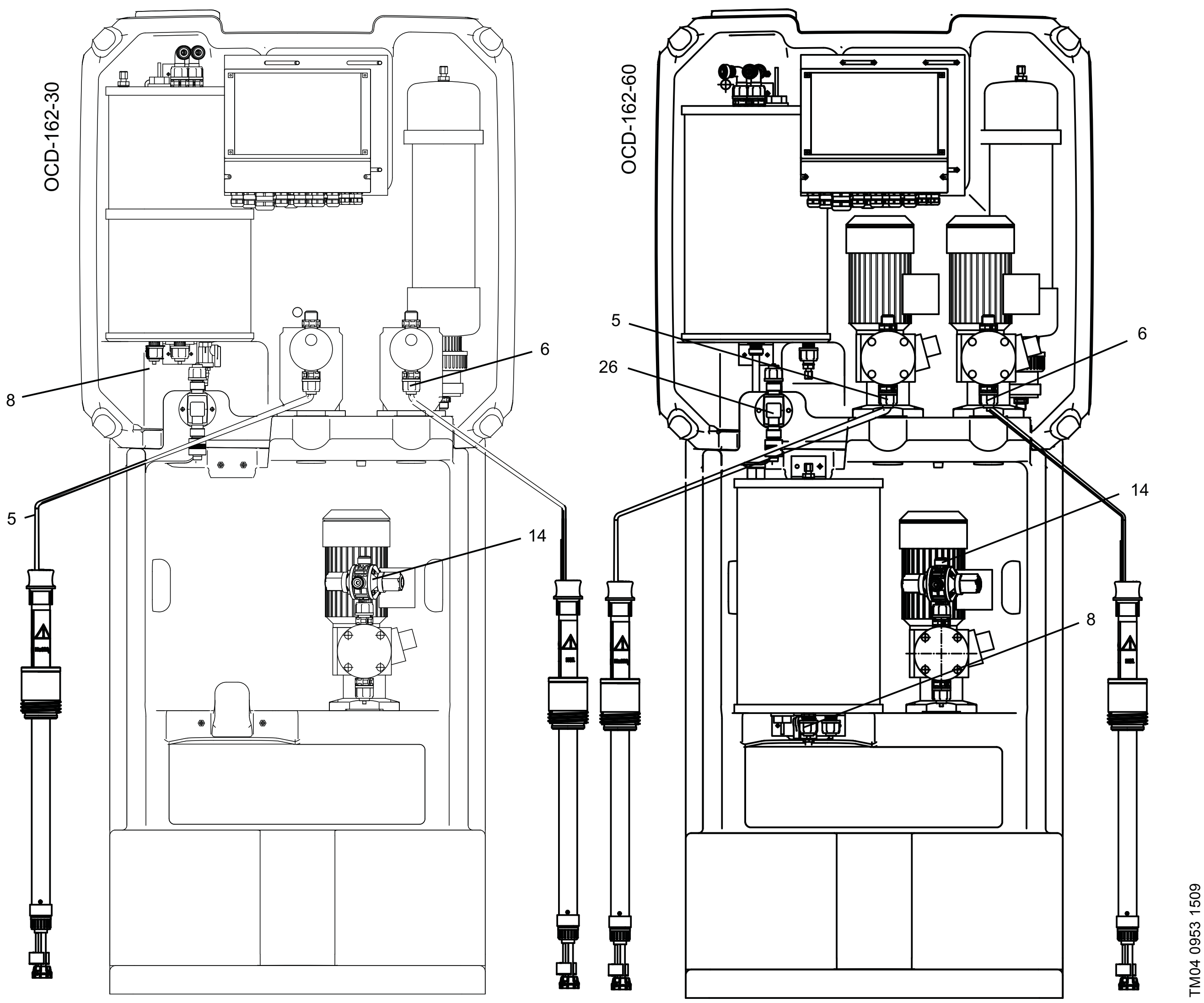

Fig. 6 Hydraulic connections OCD-162-30, -60

Pos.	Description
5, 6	Hoses for both suction lances on the suction side of the $NaClO_2$ and HCl dosing pumps
8	Hose at the drain cock of the batch tank (only installed for flushing and venting)
14	Dosing line from the dosing pump to the injection unit at the main line, or to the injection unit at the mixing module, or to the external batch tank
26	Dilution water hose at the solenoid valve

2.5.3 System connections

- Dilution water hose at the solenoid valve (26)
- Hoses for suction lances (5, 6)
- Dosing line from the dosing pump to the injection unit at the main line, or to the injection unit at the mixing module, or to the external batch tank (14)
- Hose at the drain cock of the batch tank (8)

2.5.4 External dosing pump connection

If the system has been supplied without a dosing pump, the dosing line of an external dosing pump is connected to the batch tank.

2.5.5 Measuring cell connections

The measuring cell is hydraulically connected to the main line. After dosing, the chlorine dioxide concentration, temperature and pH/ORP value of the sample water are measured in the measuring cell. The measuring cell has connections for the following:

- hose from the sample-water extraction device to the measuring cell
- hose from the measuring cell to the sample-water drain.

For more detailed information, see the installation and operating instructions for the measuring cell.

2.5.6 Measuring module connections

The measuring module is hydraulically connected to the main line. The measuring module has connections for the following:

- hose from one tapping sleeve to the measuring module
- hose from the measuring module to another tapping sleeve at the main line.

For more detailed information, see the installation and operating instructions for the measuring module.

2.5.7 Mixing module connections

The mixing module is hydraulically connected to the main line and the Oxiperm Pro system. The mixing module has connections for the following:

- dosing line from the dosing pump to the injection unit in the mixing module
- hose from one tapping sleeve to the mixing module
- hose from the mixing module to another tapping sleeve at the main line.

For more detailed information, see the installation and operating instructions for the mixing module.

English (GB)

2.6 Power supply connections and electronic connections

The Oxiperm Pro OCD-162 is equipped with an electronic controller. The controller has connections for the following:

- power supply cable to the main switch
- cable from the flow meter or contact water meter
- cable from external batch tank to level control, if applicable
- cables for measuring cell AQC-D1 or AQC-D6, if applicable:
  - measuring electrode and counter-electrode
  - sample-water sensor
  - Pt100 temperature sensor
  - pH electrode (for pH or ORP) (AQC-D1 only)
  - cleaning motor (AQC-D1 only)
- cables from the measuring module, if applicable:
  - measuring electrode and counter-electrode
  - sample-water sensor
  - Pt100 temperature sensor
- cable from the mixing module, if applicable:
  - flow controller.

For additional connections, see the separate Service Instructions.

2.7 Operation modes

During commissioning, the Oxiperm Pro OCD-162 system is set up according to the application. After switching on and starting up the chlorine dioxide production via the menu commands, the system runs fully automatically.

Two operation modes are available for the production of chlorine dioxide:

- "Internal batch tank" mode: Chlorine dioxide solution is generated in the internal batch tank, and dosed into the line system, until the batch tank is empty. There are two ways of refilling the batch tank:
  - First method "1-20": You determine how many times the batch tank is refilled by choosing a number in the range of 1 to 20.
  - Second method "0 = continuous": The batch tank is refilled continuously.
- "External batch tank" mode: Chlorine dioxide solution is generated in the internal batch tank, and dosed into an external batch tank for storage. After emptying the external batch tank, chlorine dioxide production is restarted in a continuous process.

Dosing is controlled automatically by the controller. In manual operation, the controller can be switched off.

2.8 Display and control elements

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Fig. 7 Display and control elements

Button or LED	Function
Button [Esc]	Cancels command, exits menu
Button [Up]	Selects the previous menu item, or sets a higher numerical value
Button [Down]	Selects the next menu item, or sets a lower numerical value
Button [OK]	Confirms the menu selection
Button [Cal]	Calibration
Button [Man]	Manual operation
LED "Alarm"	Alarm (red)
LED "Caution"	Warning (yellow)
LED "Cal"	Calibration (yellow)
LED "Man"	Manual operation (yellow)

2.8.1 Displays

After switching on the system, the following display appears:

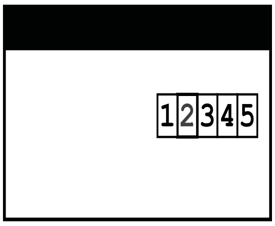

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Fig. 8 Starting the system

Press [OK] to access the "Main menu":

Main menu
Process
Controller ClO2
Alarm
Service
Setup
Maintenance

During operation, press [Esc] to access the display level:

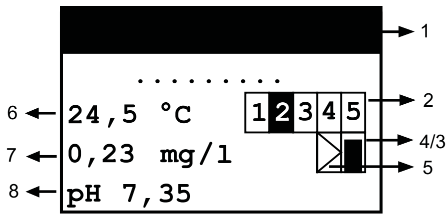

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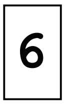
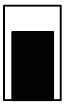
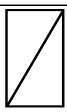
Fig. 9 "process running" display level

The header indicates the status. The positions are explained in the table below.

Legend

Pos.	Message	Description
1 Headers	"process running"	Chlorine dioxide production is active.
	"process stop"	Chlorine dioxide production has been stopped.
	"process termination"	Chlorine dioxide production has been terminated by a menu command or alarm.
	"Flushing"	Flushing is started automatically or manually.
2 Relays	1	Relay for solenoid valve: White number on a black background: relay active. Black number on a white background: relay not active.
	2	Relay for hydrochloric acid pump: display as for 1.
	3	Relay for sodium chlorite pump: display as for 1.
	4	Alarm relay: display as for 1.
	5	Warning relay: display as for 1.
3 Symbol		Symbol for the relay of the interpulse controller.
		Symbol for relay stop of the interpulse controller.
4 Symbol		Symbol for continuous controller: Box with plotted line. The height of the line is proportional to the actuating variable (chlorine dioxide dosing volume). Line not visible: actuating variable = 0 %. Line fills the entire box: actuating variable = 100 %.
		Symbol for stop of the continuous controller: White box with a diagonal line through it.
5 Symbol		Symbol for external disturbance value input (water flow as impulse/current signal). Box with plotted triangle. The black fill is proportional to the flow (the greater the fill, the greater the flow, 0-100 %). (only visible, if proportional or combined controller is configured).
6 Value	e.g. 24.5 °C	Water temperature, display value is only available with connected measuring cell.
7 Value	e.g. 0.23 mg/l	Chlorine dioxide concentration, display value is only available with connected measuring cell.
8 Value	e.g. 7.35	pH value in the sample water, display value is only available with connected measuring cell.

2.9 Access codes

When the system is ready for operation, the "Main menu" cannot be accessed without a code. Two different access authorisations or security levels are assigned for all submenus.
Each code automatically enables the levels below it.

2.9.1 Operator code

By default, all operator menus can initially be accessed without a code. When the menu selection has been confirmed with [OK], a code request is not displayed.

Once the operator has entered his operator code ("Main menu > Setup > change code"), the code request appears before any operator submenu can be accessed. The modified operator code must only allow access for operators with appropriate technical training and experience. Access is enabled for 60 minutes after entry of the code.

2.9.2 Service code

This code is reserved for trained Grundfos service engineers. Access is enabled for 30 minutes after entry. The service code is necessary for commissioning.

2.10 Menu structure

The submenus vary in submenus for operators (with operator code) and service engineers (with service code).

All software menus can be selected from the "Main menu" using the [Up] and [Down] buttons, and accessed using [OK].
Press [Esc] to return to the menu level above.

2.10.1 Operator menus (part 1)

Main menu	Submenu 1	Submenu 2	Submenu 3	Submenu 4	Submenu 5
Main menu	Process	Start	Start	Start ClO2 production?	
			Back		
		Terminate	Terminate	Terminate ClO2 production?	
			Back		
		Operation	Int. batch tank	0 = continuous	
				1-20 (adjustable)	
			Ext. batch tank	On/Off	
	Service	Process	Status	Display: process status	
			List of events		
			Production ClO2	Batches	
				Chemicals	HCl/NaClO2 reset
				Age of ClO2 (mm:ss)	
				Maintenance	
			Flushing	Start	
				Terminate	
		Measurement [1]	ClO2	Measured value	
				CalData-LogBook	
			Temperature	°C or °F (Measured value)	
			pH or ORP [5]	Measured value	
				CalData-LogBook	
		Controller ClO2			
		Water flow meter [2]	E.g. 5 impulses/sec. = 50 % or 10 mA = 50 % [3]		
		Test display			
		Program version			
Main menu	Setup	Language	Deutsch		
			English		
			(all listed)		
		Date/time	Date		
			Time		
			Daylight.sav.t.	Begin, End, time shift (± x hours), Off	
		Code function	Change		
			Delete		
		Display	Contrast		
	Alarm [4]	Alarm value ClO2	Alarm off		
			Alarm on	Alarm value 1: 0.15 mg/l	Upward violation or downward viol.
				Alarm value 2: 0.70 mg/l	Upward violation or downward viol.
				Hysteresis: 0.01	
				Alarm delay: 0 sec.	
		Dos. time monit.	Off		
			On	Max. dosing time	

1 The "Measurement" submenu only appears, if it is enabled in the "Setup" menu.

2 The "Water flow meter" submenu only appears, if a "Water flow meter" is enabled in the "Setup" menu.

3 Depending on the type of water flow meter enabled.

4 The alarm settings are only available, if "Measurement" has been enabled. The alarm relay is activated, if previously set alarm values for chlorine dioxide are exceeded or not reached, if the maximum dosing time is exceeded, and in case of a fault.

5 Depending on the setting in the "Setup" menu.

Operator menus (part 2)

Main menu	Submenu 1	Submenu 2	Submenu 3	Submenu 4	Submenu 5
calibration	chlorine dioxide	CAL meas. value			
		CAL result	slope: µA, mg/l		
		CAL cycle	On/Off		
	pH	CAL meas. value	GRUNDFOS, DIN/NIST, others		
		CAL result	Slope µA, mg/l asym. mV		
		CAL cycle	On/Off		
	ORP	CAL meas. value			
		CAL result	asym.mV		
		CAL cycle	On/Off		
manual operation	Controller ClO2	On/Off			

Operators can view the current input value of the water flow meter as well as the impulses/second, or the value in mA and the calculation in percent, see section *6.8.3*.

The value is also displayed, if the defined input values are exceeded or not reached (a water flow meter malfunction is visible here).

3. Technical data

3.1 Identification

3.1.1 Type key

Example: Oxiperm Pro OCD-162-30-D/G1

Oxiperm Pro	OCD-162	-30	-D	/G	1
Max. capacity					
5	5 g/h				
10	10 g/h				
30	30 g/h				
55	55 g/h (USA)				
60	60 g/h (USA 55 g/h)				
Chlorine dioxide dosing pump					
D	integrated mechanical dosing pump DMX				
P	integrated digital dosing pump DDI *				
S	integrated SMART Digital dosing pump DDA *				
N	without integrated dosing pump				
Supply voltage					
G	220-240 V, 50/60 Hz				
H	110-120 V, 50/60 Hz				
Suction lance					
	for 30-litre chemical container (length of suction hose 1.3 m)				
1	for 60-litre chemical container (length of suction hose 3.0 m)				
2	for 200-litre/1000-litre chemical container (length of suction hose 5.0 m)				
3	for 55-gallon chemical container (length of suction hose 3.0 m)				

* Note: It is recommended to use a digital dosing pump for direct dosing of the product solution.

3.1.2 Nameplate

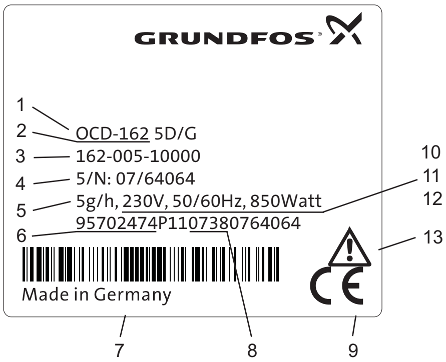


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Fig. 10 Nameplate (for OCD-162-5-D/G)

Pos.	Description
1	Type designation
2	Product name
3	Model
4	Serial number
5	Chlorine dioxide production capacity
6	Product number
7	Country of origin
8	Year and week of production
9	Marks of approval, CE mark, etc.
10	Voltage [V]
11	Frequency [Hz]
12	Power consumption
13	Safety instruction: Please read this manual

3.2 Performance and consumption

Chlorine dioxide production capacity	OCD-162-5	5 g/h
	OCD-162-10	10 g/h
	OCD-162-30	30 g/h
	OCD-162-60	60 g/h USA 55 g/h
Concentration of the chlorine dioxide solution	Approx. 2 g/l	2000 ppm
Consumption of hydrochloric acid	OCD-162-5	Approx. 0.17 l/h
	OCD-162-10	Approx. 0.37 l/h
	OCD-162-30	Approx. 0.88 l/h
	OCD-162-60	Approx. 1.71 l/h
Consumption of sodium chlorite	OCD-162-5	Approx. 0.14 l/h
	OCD-162-10	Approx. 0.30 l/h
	OCD-162-30	Approx. 0.86 l/h
	OCD-162-60	Approx. 1.63 l/h
Dilution water at 3-6 bar (drinking water quality in accordance with EU drinking water ordinance)	OCD-162-5	Approx. 2.3 l/h
	OCD-162-10	Approx. 4.8 l//h
	OCD-162-30	Approx. 14.8 l//h
	OCD-162-60:	Approx. 32.5 l//h
Max. counterpressure of chlorine dioxide dosing pump	OCD-162-5, -10	10 bar
	OCD-162-30, -60	

3.3 Temperatures and humidity

Permissible relative air humidity (no condensation)	Maximum 80 %
Permissible ambient temperature	5 °C to 35 °C
Permissible temperature of dilution water	10 °C to 30 °C
Permissible temperature of chemicals	10 °C to 35 °C
Storage temperature of the system (not connected)	-5 °C to 50 °C
Storage temperature of chemicals	5 °C to 40 °C
Permissible height above sea level, where the system is allowed to be operated.	2000 m

3.4 Dimensions

3.4.1 Oxiperm Pro OCD-162-5 and OCD-162-10

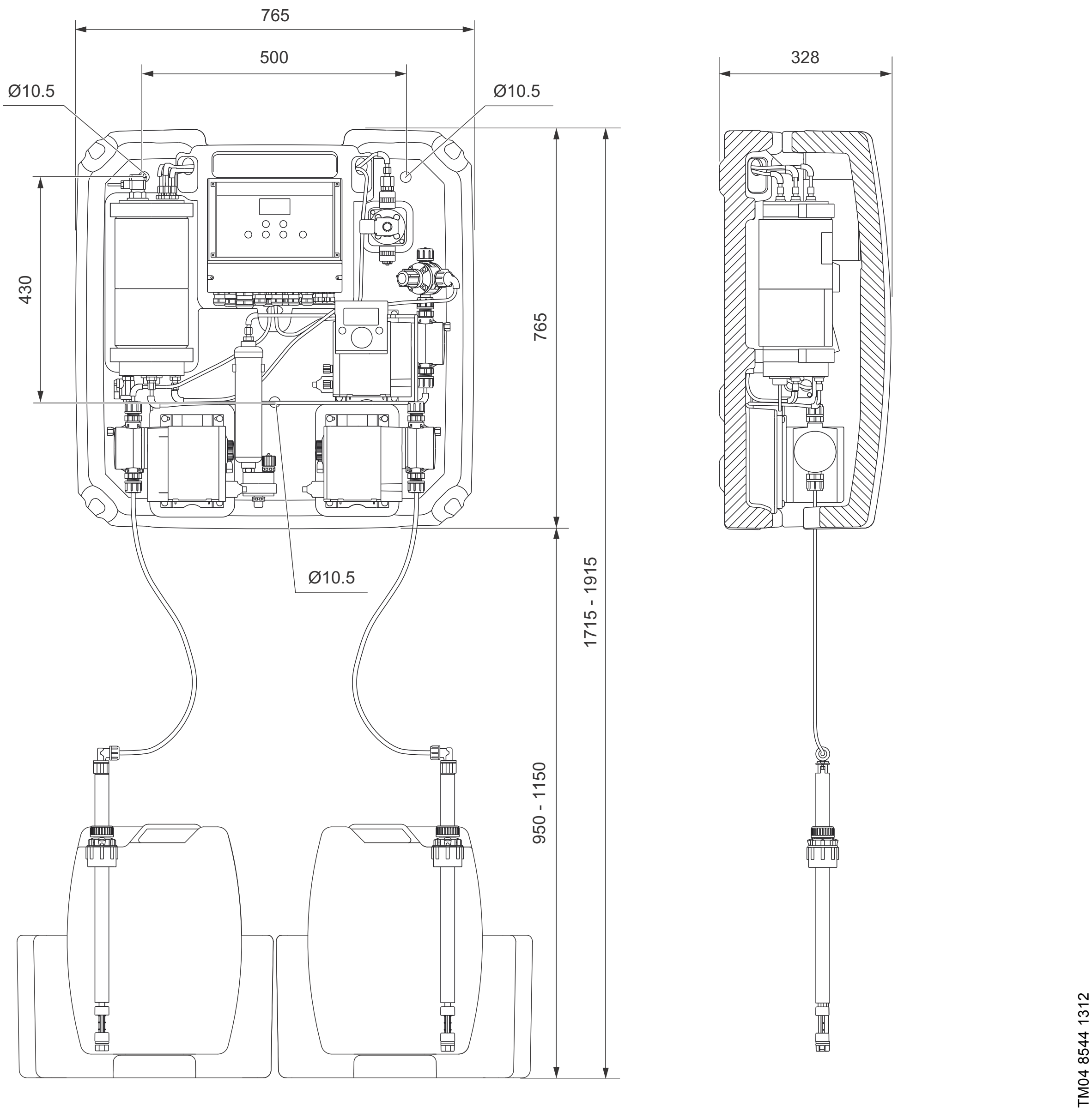


Fig. 11 Oxiperm Pro OCD-162-5 and OCD-162-10, with drill holes

3.5 Oxiperm Pro OCD-162-30 and OCD-162-60

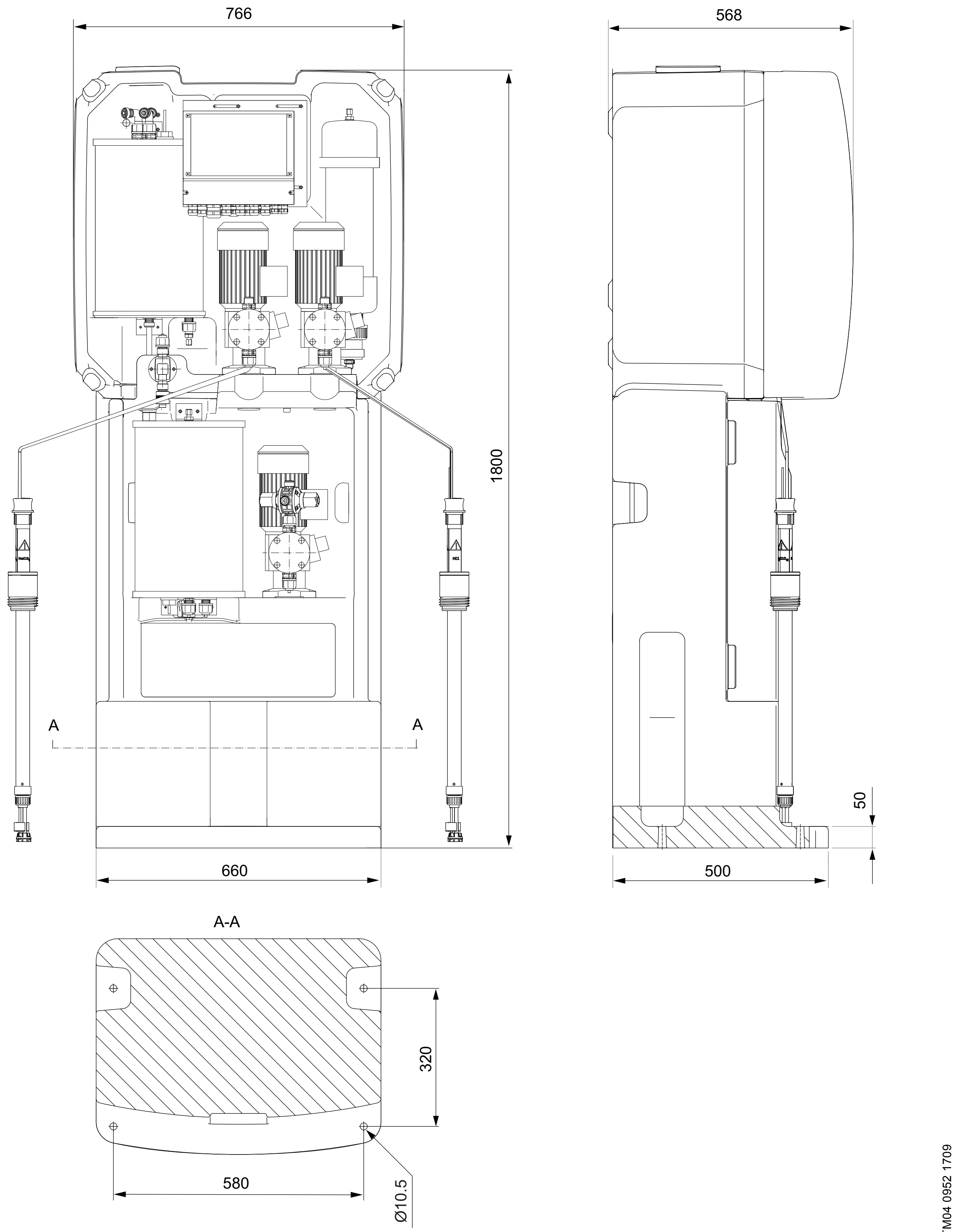


Fig. 12 Oxiperm Pro OCD-162-30 and OCD-162-60, with drill holes

3.6 Weights and capacities

System frame with cover	Oxiperm Pro OCD-162-5, -10 OCD-162-30, -60	W x H x D [mm] 765 x 766 x 330 765 x 766 x 550
Total height	OCD-162-30, -60	181 mm
Gross weight	OCD-162-5 OCD-162-10 OCD-162-30-D OCD-162-30-P OCD-162-60-D OCD-162-60-P	30 kg 32 kg 80 kg 79 kg 100 kg 99 kg
Net weight	OCD-162-5 OCD-162-10 OCD-162-30-D OCD-162-30-P OCD-162-60-D OCD-162-60-P	26 kg 28 kg 70 kg 69 kg 85 kg 84 kg
Volume compensation bag	OCD-162-5 OCD-162-10 OCD-162-30 OCD-162-60	2 l (1 unit) 4 l (2 units) 12 l (1 unit) 24 l (2 units)
Total volume of reaction tank	OCD-162-5 OCD-162-10 OCD-162-30 OCD-162-60	1.00 l 1.80 l 6.10 l 13.40 l
Total volume of batch tank	OCD-162-5 OCD-162-10 OCD-162-30 OCD-162-60	1.00 l 1.80 l 7.00 l 13.90 l
Filling volume of reaction tank	OCD-162-5 OCD-162-10 OCD-162-30 OCD-162-60	0.87 l 1.67 l 5.52 l 11.96 l
Filling volume of batch tank	OCD-162-5 OCD-162-10 OCD-162-30 OCD-162-60	0.87 l 1.67 l 6.50 l 13.00 l
Distance between bottom edge of frame and floor	OCD-162-5, -10	Approx. 1m
Total length of suction lance including suction hose	OCD-162-5, -10	1.3 m
Total length of suction lance including suction hose	OCD-162-30, -60	3.0 m or 5.0 m
Dimensions of collecting trays	Accessories	W x H x D [mm] 485 x 270 x 550
Weight of collecting trays		2 x 5.5 kg
Dimensions of external batch tank (50 l)		L x W x H [mm] 840 x 530 x 1640 diameter: 315 mm
Dimensions of external batch tank (100 l)		L x W x H [mm] 840 x 530 x 2000 diameter: 315 mm

3.7 Permissible chemicals

Nominal concentration of $NaClO_2$ solution (quality according to EN 938)	7.5 % by weight *
Nominal concentration of HCl solution (quality according to EN 939)	9.0 % by weight *

* All technical data refer to the nominal concentrations. In operation, deviations up to ± 10 % of the chemical concentrations are permissible. Yet they can change the performance data of the system specified here.

3.8 Materials

System frame	PE
Cover	EPP
Reaction tank/batch tank	PVC
Hoses	PTFE/PE
Seals	FPM/PTFE/FKM
Heads of dosing pumps	PVC

3.9 Dosing pumps

HCl/NaClO$_2$ OCD-162-5, -10 OCD-162-30 OCD-162 -60	 DDE 6-10 DDE 15-4 DMX 35-10
OCD-162-5, -10 Connection on suction side Connection on discharge side	 PE hose 4/6 PTFE hose 4/6
OCD-162-30, -60 Connection on suction side Connection on discharge side	 PVC hose 6/12 PTFE hose 9/12
ClO$_2$ OCD-162-5-P/G, -P/H OCD-162-10-P/G, -PH	DDA 7.5-16
Connection on suction side	PTFE hose 4/6
Connections on discharge side	
OCD-162-30-D/G OCD-162-30-P/G OCD-162-60-D/G OCD-162-60-P/G	DMX 16-10 DDI 60-10 DMX 35-10 DDI 60-10
Connection on suction side	PTFE hose 9/12
Connections on discharge side	

3.10 Dilution water

Extraction device for dilution water extraction	See data booklet Oxiperm Pro OCD-162
Hose connection for dilution water at the solenoid valve	Hose 6/9 mm or 6/12 or pipe PVC 10/12 mm

3.11 Main water line

Water flow meter and cable	See data booklet Oxiperm Pro OCD-162
Tapping sleeve for injection unit	

3.12 Permissible measuring cell type

AQC-D1	Measures ClO_2 + pH or ORP
AQC-D6	Measures ClO_2
Connections for sample-water extraction and drain	See data booklet Oxiperm Pro OCD-162

3.13 Product numbers OCD-162-5, -10

Product No.	OCD-162	Voltage	Dosing pump
95707848	5 g/h	230 V	DDA 7.5-16
95707849		115 V	
95702476		230 V	Without dosing pump
95702477		115 V	
95707850	10 g/h	230 V	DDA 7.5-16
95707851		115 V	
95702480		230 V	Without dosing pump
95702481		115 V	

All dosing pumps are applicable for 50/60 Hz.

3.14 Product numbers OCD-162-30, -60

Product No.	OCD-162	Voltage	Suction hose	Dosing pump
95718444	30 g/h	230 V	3.0 m	DMX 15-10
95718445		115 V		
95718446		230 V		DDI 60-10
95718447		115 V		
95718448		230 V	5.0 m	DMX 15-10
95718449		115 V		
95718450		230 V		DDI 60-10
95718451		115 V		
95718452	60 g/h USA 55 g/h	230 V	3.0 m	DMX 35-10
95718453		115 V		
95718454		230 V		DDI 60-10
95718455		115 V		
95718456		230 V	5.0 m	DMX 35-10
95718457		115 V		
95718458		230 V		DDI 60-10
95718459		115 V		

All dosing pumps are applicable for 50/60 Hz

3.15 Electrical data

Power supply connection		115 V or 230 V, 50/60 Hz
Input power of the basic system without external consumer load	OCD-162-5, -10	Max. 100 VA
	OCD-162-30	Max. 180 VA
	OCD-162-60	Max. 320 VA
Input power of the whole system		Max. 850 VA
Maximum permissible load on the potential-free output contacts		Max. 550 VA (250 V x 2 A)
Enclosure class of electronics		IP65
Enclosure class of dosing pump		
Enclosure class of solenoid valve		

3.16 Controller inputs

Input	Description
Analog input for flow meter	Current input 0(4)-20 mA Load: 50 Ω
Analog input	Chlorine dioxide concentration Measuring cell (optional)
	Pt100 temperature sensor in the measuring cell
Contact input (compound-loop control)	Contact water meter Maximum 50 impulses/second Maximum voltage: 13 V
External input: stop	For process enabling and for external fault
mV input	pH or ORP
53, 54, H_2O	Sample-water sensor input for the measuring cell Maximum voltage: 13 V

Switching input	Description
K1	Reaction tank: water supply to level K1
K2	Reaction tank: HCl supply level
K3	Reaction tank: $NaClO_2$ supply level
K4	Reaction tank: water supply to level K4
K5	Batch tank level Empty signal
K6	Batch tank level Maximum level
K7	HCl container level: low-level signal Contact open: low-level signal HCl
K8	HCl container level: Empty signal Contact open: empty signal HCl
K9	$NaClO_2$ container level: low-level signal Contact open: low-level signal $NaClO_2$
K10	NaClO2 container level: empty signal Contact open: empty signal $NaClO_2$
K11	External batch tank: min. level
K12	External batch tank: max. level
K13	External batch tank: max. -max. level

3.17 Controller outputs

Outputs	Description
Analog output 0(4)-20 mA	Current output Control
Analog output for external device (proportional to ClO_2 concentration)	Current output Measured value for check measurements 0(4)-20 mA Load: 500 Ω
Solenoid valve for water supply	Relay 1
Dosing pump for HCl	Relay 2
Dosing pump for $NaClO_2$	Relay 3
Alarm relay (changeover contact) Potential-free output	Relay 4
Warning relay Potential-free output	Relay 5
Dosing pump for ClO_2	Relay 6

4. Transport and packaging

Warning

Increased risk of damage to equipment and personal injury as a result of operating faults due to transport damage.

Do not shake, crush or drop the box.
Do not use a sharp or pointed blade. Open the packaging carefully.

Carefully remove the device from the box.
Do not bend the hoses and cables.

Note ***Do not adjust the stroke-length adjustment knob on the pump. It must not be adjusted until the pump is running.***

4.1 Unpacking

Number of packing units: 1 box.

Dimensions L x W x H [mm]	Contents	Weight gross/net [kg]
900 x 900 x 518	Oxiperm Pro with cover, hoses, screws, accessories	OCD-162-5: 30/26 OCD-162-10: 32/28
766 x 558 x 1813		OCD-162-30-D: 80/70 OCD-162-30-P: 79/69
766 x 558 x 1813		OCD-162-60-D: 100/85 OCD-162-60-P: 99/84

1. Unpack the device.
2. Unpack the cover.
3. Unpack the measuring cell, if supplied.
4. Unpack the extension modules, if supplied.
5. Retain the original packaging in order to return the device for servicing.
6. Check the device(s) for transport damage (especially hoses and cables).

4.2 Transport damages

1. Pack the device in its original packaging.
2. Inform the forwarder of the transport damage.
3. Return the device to the supplier.

5. Installation

Installation is described in detail in the separate Service Instructions. This subsection can be used by the operating company to plan installation.

5.1 Preparing the installation location

The operating company must ensure that all the conditions listed below for structurally and technically safe and optimum operation of the system are met prior to commencing installation.

The installation location must fulfil the following:

- It is protected from the sun, frost-proof, well-ventilated and has sufficient lighting. The Oxiperm Pro system must not be installed outdoors.
- It meets the conditions specified in section *8. Accessories list* regarding air temperature, humidity, permissible component temperature and dilution water quality.
- It has concrete walls and floors, which enable the OCD-162-5, -10 system to be wall-mounted (minimum wall thickness for the mounting screws 100 mm), or the OCD-162-30, -60 system to be floor-mounted.
- It has a power supply connection, see section *8.*
- It has access to the main water line.
- It has a connection for dilution water of drinking water quality with a isolating valve.
- It has a floor drain for washing away chemicals and a drain (tank) for sample water.
- It has a separate storage room for empty and full chemical containers.
- It is isolated from other areas with regard to fire protection.
- It is secured against unauthorised access, and meets the regulations for the prevention of accidents.
- It is not in permanent use by personnel. The maximum stay is two hours.

Note ***We recommend the installation of a gas warning device.***

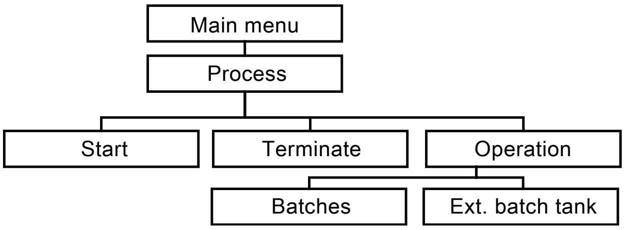
5.2 Preparing for installation

Checklist	Done
1. Read the installation and operating instructions for the Oxiperm Pro system, the dosing pumps, the multi-function valve, the measuring cell and the extension module, if used. Store the manuals in a dry place in the installation location.	
2. Measure the pressure and temperature in the dilution water line and in the main water line.	
3. Measure the room temperature and humidity.	
4. Obtain official approval for storing chemicals, if necessary.	
5. Purchase accessories, see section *8.*	
6. Fit a tapping sleeve for the dilution water supply in the drinking water line.	
7. Fit a tapping sleeve for the injection unit in the main line.	
8. Install a protective pipe for the dosing line, if necessary.	
9. Fit a tapping sleeve for sample-water extraction at the main line, if necessary.	
10. Fit tapping sleeves for the measuring or mixing module, if used.	
11. Provide protective clothing in the room in accordance with the regulations for the prevention of accidents (Germany GUV-V D5).	
12. Display all warning signs provided. Display a warning sign: "No fires, naked flames or smoking".	

6. Operation

All software menus can be selected from the "Main menu" using the [Up] and [Down] buttons, and accessed using [OK]. Press [Esc] to return to the previous menu level.

The menu structures for the submenu levels are illustrated by means of a graphic.



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Structure of the "Process" menu

6.1 Switching on the system

1. Open the dilution water supply.
2. Switch on the main switch for power supply.

The system starts.

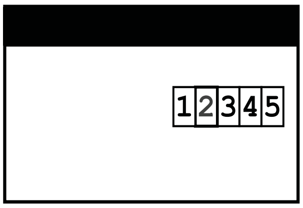

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Fig. 13 Start-up display

3. Press [OK] to confirm. The "Main menu" appears. The system is ready for use.

Main menu
Process
Controller ClO2 *
Alarm
Service
Setup
Maintenance

* The "Controller ClO2" submenu only appears, if the controller has been defined under "Setup" > "Controller ClO2".

6.1.1 Adjusting the process (how many batches)

To specify whether just one batch or several batches of chlorine dioxide is to be prepared, or whether the production process should run continuously, proceed as follows:

1. "Main menu > Process" > [OK].

Process
Start
Terminate
Operation

"Operation" > [OK]. The factory setting is "Int. batch tank"

Operation
Int. batch tank
Ext. batch tank

"Int. batch tank" > [OK], the factory setting is confirmed.

Int. batch tank
X Batches
0 = continuous

The system produces the chosen number of batches.
The setting "0 = continuous" means continuous production.

2. Or "Ext. batch tank" > [OK]. The system is now set to "Ext. batch tank" operation.

In the operation modes "0 = continuous" or "Ext. batch tank", anew production process is started when the batch tank is empty.

In the operation mode "Batches", the chlorine dioxide production stops when the reaction tank is empty.

6.2 Starting operation

6.2.1 Starting chlorine dioxide production

1. "Main menu > Process" > [OK].

Process
Start
Terminate
Operation

2. "Start" > [OK].

Start
Start
Back

3. "Back" > [OK], the command is cancelled.
4. "Start" > [OK], the command is executed.

Start
Start ClO2 production?

5. Press [OK], chlorine dioxide production starts.

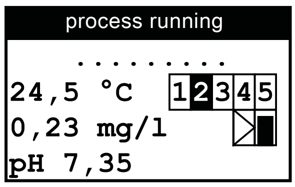

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Fig. 14 Display level during operation

Dosing starts automatically as soon as the batch tank is filled.

6.3 Interrupting operation

6.3.1 Terminating chlorine dioxide production

1. "Main menu > Process" > [OK].

Process
Start
Terminate
Operation

2. "Terminate" > [OK].

Process
Terminate
Back

3. "Back" > [OK], the command is cancelled.
4. "Terminate" > [OK], chlorine dioxide production is terminated.

process stop
Terminate ClO2 production?

5. Press [OK]. The chemical dosing pumps are stopped.
   The reaction tank is filled with water up to level K4, in order to dilute the undefined content of the reaction tank.

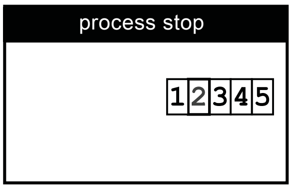

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Fig. 15 Display level process stop

Note

The chemical reaction in the reaction tank continues. The dosing pump continues running, until the chlorine dioxide solution is emptied from the batch tank.

For continuing operation after terminating chlorine dioxide production, see 6.4.1.

6.3.2 Terminating the dosing process

- Switch off the controller in normal operation. The dosing pump stops.
- It is not recommended to switch off the controller in "Setup", because all the controller parameters will be lost for resuming the dosing process. For details, see *6.3.3*.
- For switching off the controller in manual operation, see *6.14*.

6.3.3 Resuming the dosing process

- Switch on the controller in normal operation.

6.4 Continuing operation after an interruption

6.4.1 Continuing operation after terminating chlorine dioxide production

During commissioning and in normal operation: Start chlorine dioxide production:

– "Main menu > Process" > [OK].
– "Start" > [OK].
– "Start" > [OK].

The undefined content of the reaction tank is flushed into the batch tank, and the warning message "check ClO2 batch" appears. The batch should be disposed, see section *6.5.2*.

After that, the process and dosing proceed in normal operation.

6.4.2 Continuing operation after removing a fault

Confirm the alarm message. Continue operation as described in section *6.4.1*.

6.4.3 Continuing operation after changing a chemical container

The Oxiperm Pro system automatically continues operation.

6.4.4 Continuing operation after a power supply interruption

As soon as the power is restored, the Oxiperm Pro system is switched on again automatically.

If the reaction tank is full, the liquid is in an undefined state. It can contain too much hydrochloric acid or too little $NaClO_2$. The controller uses the float switch in the reaction tank to determine whether the reaction tank is empty or full, and continues operation accordingly:

- The reaction tank is part-filled or full, the batch tank is empty:
  – If the reaction tank is part-filled, it is filled up with water. The undefined liquid is drained into the batch tank, and the following alarm message is displayed: "check ClO2 batch".
  – Drain the batch tank manually. For details, see section *6.5.2*. If it is not emptied manually, the undefined liquid will be dosed.
  – The dosing pump is switched off when the batch tank is empty.
- The reaction tank is empty, the batch tank is part-filled or full:
  – In continuous ClO_2 production or operation with external batch tank, a new production process will be started.
- The reaction tank is part-filled or full, the batch is tank part-filled or full:
  – The remaining batch is dosed into the batch tank.
  – The reaction tank is filled up with water. The undefined liquid remains in the reaction tank, until the batch tank is empty. Water is supplied to make the reaction tank overflow, and the alarm message "check ClO2 batch" is displayed.
  – Drain the batch tank manually. If it is not drained manually, the undefined solution will be dosed.
  – In continuous ClO_2 production a new production process is started after the reaction tank is empty.

6.5 Flushing

Warning

Risk of poisoning from chlorine dioxide gas.

Do not mix sodium chlorite and hydrochloric acid.

Do not put suction lances in the same bucket.

Do not insert suction lances in the wrong container.

Observe the labels on chemical containers, suction lances and pumps:
red = HCl, blue = $NaClO_2$.

Warning

Risk of serious personal injury and damage to equipment and due to incorrect handling of chemicals.

Before starting work, put on protective clothing (gloves, face mask, protective apron).

Warning

Risk of burns from stray droplets when removing the suction lance from the chemical container. Take care that no stray droplets fall on skin, clothing, shoes or the floor.

Any stray droplets on the container or in the collecting tray must be immediately rinsed away with water.

The Oxiperm Pro system has two flushing functions:

- Automatic flushing, see section *6.4.4*
- Flushing with the "Flushing" menu command, described in this section.

The Oxiperm Pro system must be flushed with the "Flushing" menu command in the following situations:

- prior to an extended shutdown of the system
- prior to maintenance and servicing work.

The entire system is flushed with water to remove any traces of chemicals in the suction hoses, pumps and reaction tank. Dilution water is used for flushing. Before flushing, the suction lances have to be removed from the chemical containers, and placed in individual 10-litre buckets with drinking water so that they take up the water. Each pump is flushed for four minutes.

Before flushing, the batch tank must be manually drained via the drain cock.

- Only start flushing when the chlorine dioxide production is not running. Otherwise the "Flushing" menu is not visible.
  The flushing process can be stopped anytime with [Esc].

6.5.1 Preparing to flush

1. Before starting to flush, have ready the following items:
   – Empty 10-litre plastic buckets (1 bucket for OCD-162-5, -10, -30, and 2 buckets for OCD-162-60)
   – PE hose, 11 x 8 mm, for the drain cock on the batch tank
   – Chlorine dioxide breakdown substance (sodium thiosulfate $Na_2S_2O_3$ x 5 H_2O): 20 g for OCD-162-5, 40 g for OCD-162-10, 120 g for OCD-162-30, 240 g for OCD-162-60
   – Two 10-litre buckets filled with water
   – The original screw caps for the chemical containers.

6.5.2 Manually draining the batch tank

1. Place the two 10-litre buckets filled with water to the right and left of the chemical containers.
2. Untwist the screw cap on the suction lance of the sodium chlorite container. Remove the suction lance, and place it in one of the buckets of water.
3. Screw the original cover onto the sodium chlorite container.
4. Untwist the screw cap on the suction lance of the hydrochloric acid container. Remove the suction lance, and place it in the other bucket of water.
5. Screw the original cover onto the hydrochloric acid container.
6. Fill the empty bucket with 1 litre of water and 20 g (OCD-162-5), 40 g (OCD-162-10) or 120 g (OCD-162-30) of chlorine dioxide breakdown substance, and place it to the left of the Oxiperm Pro. For the OCD-162-60, fill two empty buckets, each with 1 litre of water and 120 g of chlorine dioxide breakdown substance, and place them to the left of the Oxiperm Pro.
7. Remove the cover from the Oxiperm Pro.
8. Connect one end of the hose (PVC) to the drain cock of the batch tank, and place the other end in the bucket. Open the drain cock.
9. Empty the contents of the batch tank (OCD-162-5 around 1 litre, OCD-162-10 around 1.8 litres, OCD-162-30 around 6.5 litres, OCD-162-60 around 13.0 litres) into the bucket(s).
10. When the batch tank is empty, close the drain cock.

6.5.3 Starting flushing

1. Main menu > Service" > [OK].
2. "Process" > [OK].
3. "Flushing" > [OK].

Start flushing:

4. "Start" > [OK].

Flushing
Start system flushing?

5. Press [OK].

Flushing
Put suct. lance into water [OK]

6. Press [OK], if the suction lances are in the water.

Flushing
Drain int. batch tank [OK]

7. Press [OK], if the batch tank is empty, see section *6.5.2 Manually draining the batch tank*.

Flushing
Close drain cock [OK]

8. Press [OK], if the drain cock is closed.
9. Flushing starts. The flushing process automatically runs twice.

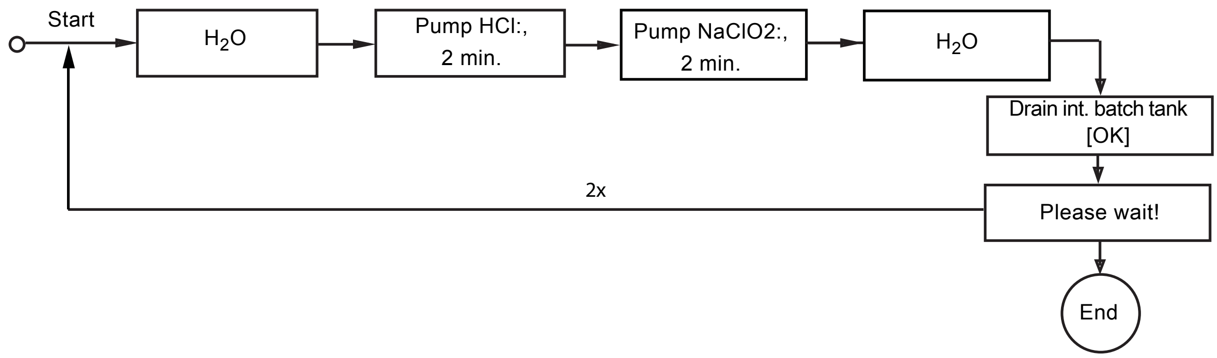

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Fig. 16 Flushing process

10. Flushing process
    – Dilution water is filled up in the reaction tank until the level reaches K1.
    – The hydrochloric acid pump runs for two minutes.
    – The sodium chlorite pump runs for two minutes.
    – Dilution water is filled up in the reaction tank until the level reaches K4, and it overflows into the batch tank.
    The following message is displayed:

Flushing
System flushing is running

At the end of flushing, the following message is displayed:

Flushing
Please wait!

11. If the reaction tank is empty, the flushing process is repeated. After two flushing processes, the following message is displayed:

Flushing
System flushing is finished [Esc]

12. Confirm with [OK].

6.5.4 Working with the system after flushing

1. Remove the hose from the drain cock, and place it in the bucket.
2. Pour the contents of the bucket down the drain.
3. Clean the hose and bucket, and drain thoroughly with water.
4. Unscrew the original cover of the sodium chlorite container.
5. Remove the sodium chlorite suction lance from the bucket of water, and insert it in the sodium chlorite container. Screw the suction lance cap onto the container. Retain the original cover.
6. Unscrew the original screw cover of the hydrochloric acid container.
7. Remove the hydrochloric acid suction lance from the bucket of water, and insert it in the hydrochloric acid container.
   Screw the suction lance cap onto the container. Retain the original cover.
8. Fit the cover back on the Oxiperm Pro.

6.5.5 Terminating flushing

The flushing process can be terminated any time:

1. Main menu > Service > Process > Flushing > [OK].

Flushing
Start
Terminate

2. Terminate > [OK].

Flushing
start again
Terminate

Terminate flushing:

3. Terminate > [OK].

Restart flushing:

4. start again > [OK].

6.5.6 Responding to flushing faults

It can take five minutes to rinse the pumps. If the following error message appears after five minutes, proceed as follows:

Flushing
flushing error

1. Call Service. There may be a problem with the pumps.
2. Once the fault has been removed, press [Esc] or [OK] to confirm.
3. Restart flushing.

6.6 Manually venting the dosing pump

Warning

Risk of damage to equipment and personal injury due to incorrect handling of chemicals.

Before commencing work, put on protective clothing.

Note

The dosing pump can be vented manually with the multi-function valve.

Only actuate the green knob on the multi-function valve when the dosing pump is running.

Conditions for venting:

- The batch tank must contain chlorine dioxide solution.
- The dosing pump must be operated in "manual" mode.

If the dosing pump is running:

1. Hold the green knob and the black knob on the multi-function valve. Turn the green knob slightly clockwise as far as the stop (the movement is barely noticeable). For details, see installation and operating instructions for the multi-function valve.
2. Repeat the procedure, if necessary.

The dosing pump is vented.

Turning the knob opens the pressure loading valve at the multi-function valve to the overflow line. The chlorine dioxide solution, possibly containing air bubbles, flows back to the batch tank.

6.7 Modifying setup

Setup is selected as follows:

1. Main menu > Setup > [OK].

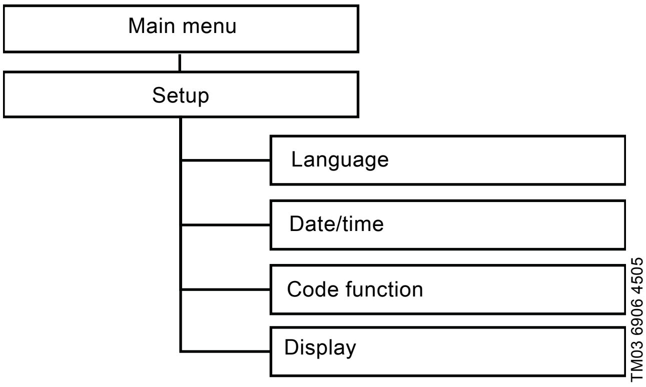

Fig. 17 Structure of the "Setup" menu for operators

Note ***Other menus can only be accessed with the service code.***

6.7.1 Changing the language

1. "Main menu > Setup" > [OK].
2. "Language" > [OK].
3. Select the operator language > [OK] > [Esc].

6.7.2 Changing the operator code

1. "Setup" > "Code function" > [OK].

Code function
change code
delete code

2. "change code" > [OK].

A request for the old code appears:

Code function
old code: 0

3. Press the [Up] button to set the old code, confirm with [OK].

Code function
new code: 0

4. Press the [Up] button to set the new code (maximum 9999), confirm with [OK].

The new code is set. The code request now appears before all submenus that can be accessed by operators.

Delete code

1. "delete code" > [OK].

The code setting is deleted. Access is now only possible with the factory-set operator code 0000.

6.7.3 Setting the time

1. "Main menu > Setup" > [OK].
2. "Date/time" > [OK].
3. "Time" > [OK].
4. Press the [Up] or [Down] button to set the commissioning time, confirm with [OK].

The settings are saved.

6.7.4 Setting the date

1. "Main menu > Setup" > [OK].
2. "Date/time" > [OK].
3. "Date" > [OK]. The current date is displayed.
4. Press [OK] to confirm, or use the [Up] or [Down] button to set the commissioning date. Confirm with [OK].

6.7.5 Setting or cancelling daylight saving time

1. "Main menu > Setup" > [OK].
2. "Daylight.sav.t.."> [OK].
3. Use the [Up] or [Down] button to make your settings.

6.7.6 Changing the display contrast

1. "Main menu > Setup" > [OK].
2. "Display" > [OK].

The display contrast is shown in percent.

3. Press [Up] to increase the contrast, press [Down] to reduce the contrast. Confirm with [OK].

The settings are saved.

6.7.7 Program version

1. "Main menu > Service > Program version" > [OK].

Program version
Oxiperm Pro 162
5 g/h
V 0.20.020090426

2. View program version > [Esc]

6.8 Monitoring the production and dosing process

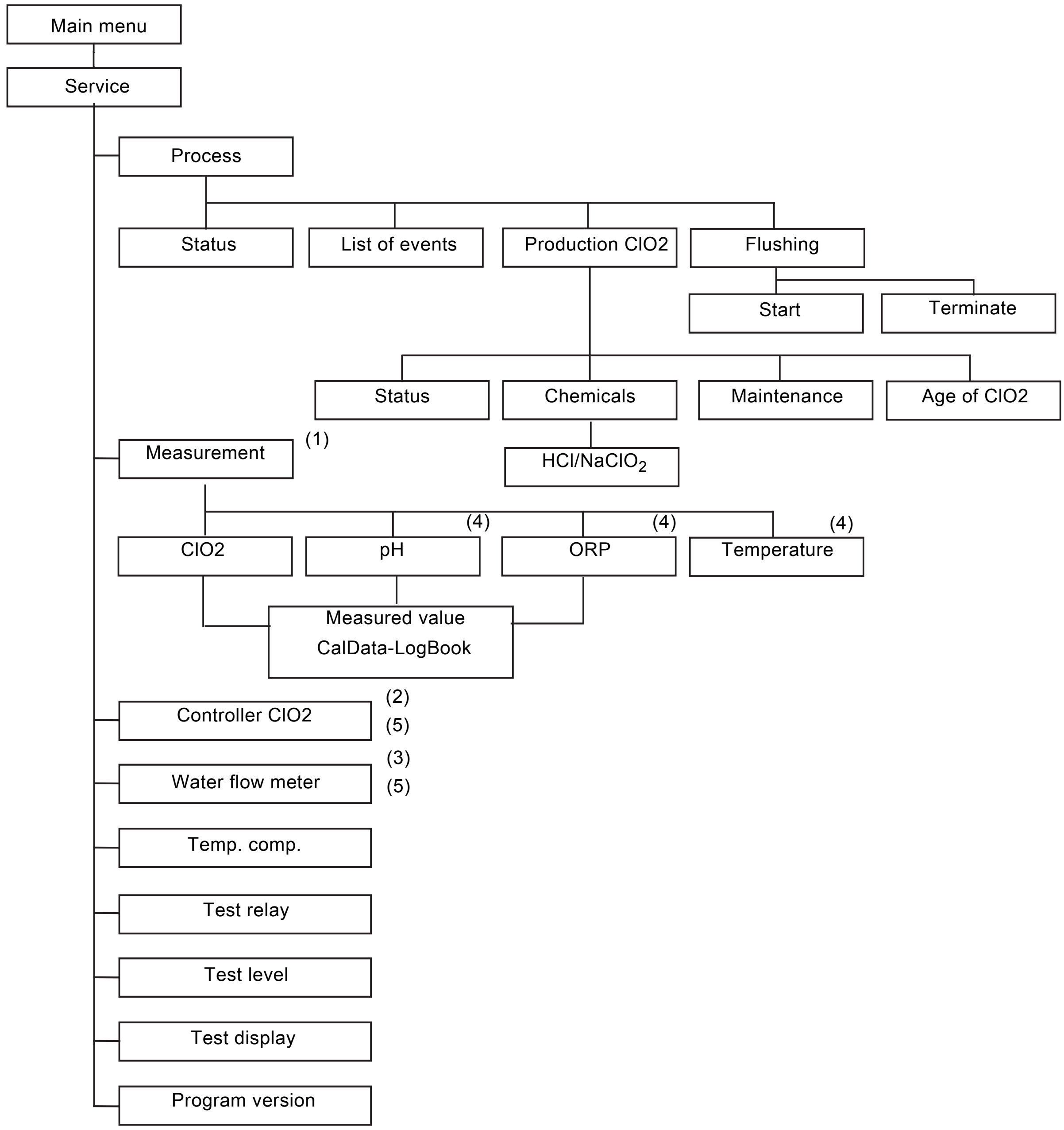

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Fig. 18 Structure of the "Service" menu for service engineers

(1) Only visible, if "Measurement" is activated.

(2) Only visible, if "Controller ClO2" is activated.

(3) Only visible, if "Water flow meter" is activated.

(4) Different submenus, according to the selected measuring cell and measured variable.

(5) Not visible for applications with external batch tank.

6.8.1 Current measured values for ClO_2, water temperature and pH/ORP

Note

The measured values in this menu are only displayed, if "Measurement" has been enabled.

If the AQC-D6 measuring cell has been selected, only the ClO_2 measured value is displayed.

If the AQC-D1 measuring cell has been selected, the ClO_2 and the pH/ORP measured values are displayed.

1. "Main menu > Service" > [OK].
2. "Measurement" > [OK].

Measurement
ClO2
Temperature
pH/ORP

ClO2 measured value

1. "ClO2" > [OK].

measurement ClO2
CalData-LogBook
Measured value

2. Press [Down] to select "Measured value", confirm with [OK].

The current ClO_2 measured value is displayed:

– ClO_2 concentration [mg/l],
– current from measuring cell [µA]),
– set measuring range [mg/l)].

measurement ClO2
0.00 mg/l
0.000 µA
0.0 - 1.0 mg/l

Calibration logbook for ClO2

1. Press [Up] to select "CalData-LogBook", confirm with [OK].

Measurement
ClO2
Temperature
pH

2. ClO2 > [OK]. The last 10 calibration data records are listed in chronological order in "CalData-LogBook". Data record 1 is the one that was last saved.
3. Press [OK] to display the previous data record.

CalData-LogBook	
number	1
Date	2008-09-14
Time	7:54
slope:	20.2

pH value

1. "Main menu > Service" > [OK].
2. "Measurement" > [OK].

Measurement
ClO2
Temperature
pH

3. pH > [OK].

Measurement pH
CalData-LogBook
Measured value

4. Press [Down] to select "Measured value", confirm with [OK].

The current pH measured value is displayed:

– pH measured value,
– voltage [mV],
– set measuring range [pH].

Measurement pH
7.20 pH
-30 mV
0.00 - 14.00 pH

Calibration logbook for pH

1. Press [Up] to select "CalData-LogBook", confirm with [OK].

Measurement
ClO2
pH

2. pH > [OK]. The last 10 calibration data records are listed in chronological order. Data record 1 is the one that was last saved.
3. Press [OK] to display the previous data record. Press [Up] or [Down] to display all lines with calibration data.

CalData-LogBook	
number	1
Date	2008-09-23
Time	09:01
slope:	-54.2
asym.	11.31
buffer 1	4.01
buffer 2	7.00
Cal temp.	25.0

ORP value

1. "Main menu > Service" > [OK].
2. "Measurement" > [OK].

Measurement
ClO2
ORP

3. "ORP" > [OK].

MeasurementORP
CalData-LogBook
Measured value

4. Press [Down] to select "Measured value", confirm with [OK].

MeasurementORP
-1600 mV
-1500 mV

The current ORP measured value is displayed:
  – ORP measured value: voltage in [mV],
  – set measuring range in [mV].

5. Press [Esc].

Calibration logbook for ORP

1. Press [Up] to select "CalData-LogBook" > [OK].

Measurement
ClO2
ORP

2. "ORP" > [OK]. The last 10 calibration data records are listed in chronological order in the "CalData-LogBook". Data record 1 is the one that was last saved.
3. Press [OK] to display the previous data record.

CalData-LogBook	
number	1
Date	2008-09-23
Time	08:54
offset:	-4.49

Temperature of the sample water

1. "Main menu > Service" > [OK].
2. "Measurement" > [OK].
3. "Temperature" > [OK].
4. "Measured value" > [OK].

The current measured value is displayed:
  – temperature [°C],
  – set measuring range.

If the measuring range is exceeded or not reached, a fault has occurred (for example temperature sensor cable breakage).

Temperature
23 °C
0.0 - 50.0 °C

6.8.2 Current control parameters

Not for applications with external batch tank.

Parameters for the setpoint controller

1. "Main menu > Service" > [OK].
2. "Controller ClO2" > [OK].

Controller ClO2
y-out: %:............. 75 %
setpt: mg/l:....... 0.40 mg/l
setpoint ctrl.
XP:............................. 83 %
TN:......................... 300 sec.
(TV)
dosing capacity:......... 100 %

Menu	Description	Visible for
y-out: %	Controller output signal to the dosing pump	
setpt: mg/l	Setpoint in mg/l	
setpoint ctrl.	Controller type	
XP	Proportional band: When selecting the P controller range, the actuating variable (dosing volume) is proportional to the system deviation (difference between actual value and setpoint)	
TN	Reset time	P, PI controller, PID
(TV)	Derivative action time	PID controller
Qmax: %	Maximum dosing flow (0-100 %) (the value entered in the "Controller ClO2" menu under "dosing flow")	
minOn: s	Minimum on-time	Interpulse controller

Parameters for the proportional controller

1. "Main menu > Service" > [OK].
2. "Controller ClO2" > [OK].

Controller ClO2
y-out: %:............... 75 %
addit.: mg/l:........ 0.40 mg/l
proport. ctrl.
Qmax: %:........... 100 %
minOn: s:......... 1.0 sec.
dos.coeff.:....................... 1.0

Menu	Description	Visible for
y-out: %	Controller output signal to the dosing pump.	
addit.: mg/l	0.20 mg/l	
proport. ctrl.	Controller type	
Qmax: %	Maximum dosing flow (0-100 %) (the value entered in the "Controller ClO2" menu under "dosing flow")	
minOn: s	Minimum on-time	Interpulse controller
dos.coeff.:	Dosing coefficient, calculated by the controller	

6.8.3 Current input value for water flow meter

Not for applications with external batch tank.

1. "Main menu > Service" > [OK].
2. "Water flow meter" > [OK].

Contact water meter

Water flow meter
1.20 imp./sec.
54 %

The display shows:

- The rate of impulses from the contact water meter in impulses per second.
- The percentage of the entered maximum flow (for example 40 m^3/h => 54 % x 40 m^3/h = 21.6 m^3/h).

If the defined input values are exceeded or not reached, a fault has occurred, for example at the water flow meter.

Flow meter

Water flow meter
10 mA
54 %

The display shows:

- The current in mA corresponding to the flow.
- The percentage of the entered flow (for example $Q_{max.}$ = 40 m^3/h => 54 % = 21.6 m^3/h).

6.8.4 Process status

1. "Main menu > Service" > [OK].
2. "Process" > [OK].
3. "Status" > [OK].

process status
HCl supply running

Examples of status messages:

Status message	Description
H2O supply 1 running	Start of ClO_2 production, dilution water 1, relay 1
HCl supply running	Pump HCl, relay 2
NaClO2 supply running	Pump NaClO2:, relay 3
reaction time	Timer running, showing the remaining reaction time
H2O supply 2 running	Dilution water 2, relay 1
Filling int. batch tank	After dilution water 3, relay 1
system waiting	Waiting until the batch tank is empty
process stop	An alarm message has caused the process to stop
process termination	An alarm message or menu command has caused the process to be terminated

6.8.5 List of events

The list of events is useful for fault finding. Faults and messages are stored chronologically in the list of events.

1. "Main menu > Service > Process" > [OK].
2. "List of events" > [OK]. Record number 1 is the most recently stored, record number 20 is the oldest. The oldest record is deleted when a new one is stored.
   – Press [Down] to scroll through the list. For details of possible events, see the tables with alarm messages in section *6.11*.

List of events
number......................... 1/99
process termination
2008-07-22................. 11:45

6.8.6 Number of ClO2 batches

1. "Main menu > Service > [OK].
2. "Production ClO2" > [OK].
3. "Batches" > [OK].

After 9999 batches, the display is reset to 0.

Production ClO2
25
Batches

6.8.7 Maintenance date

1. "Main menu > Service" > [OK].
2. "Production ClO2" > [OK].
3. "Maintenance" > [OK].

Maintenance
LAST
2012-07-25
NEXT
2013-07-25

6.8.8 Resetting the chemical consumption after changing containers

The controller calculates the chemical consumption and displays it in litres. It starts automatically at 0.000 litres.

Displaying the chemical consumption

1. "Main menu > Service" > [OK].
2. "Process" > [OK].
3. "Production ClO2" > [OK].
4. "Chemicals" > [OK].

Chemicals
HCl
NaClO2
reset

5. HCl> [OK].

HCl
0.000 L
since 2012-04-29

6. Press [Esc].
7. "NaClO2" > [OK].

Resetting the chemical consumption

1. "Production ClO2" > [OK].
2. "Chemicals" > [OK].
3. "reset" > [OK].

reset
HCl
NaClO2

4. HCl > [OK]. The counter is reset to 0.
5. NaClO2 > [OK]. The counter is reset to 0.

6.8.9 Indicating the "Age of ClO2" in the reaction tank and in batch tank

1. "Main menu > Service > [OK].
2. "Production ClO2" > [OK].

Production ClO2
Batches
Chemicals
Maintenance
Age of ClO2

3. "Age of ClO2" > [OK]. The factory setting for both is 00:00 (minutes and seconds).

Age of ClO2
reaction tank
03:16
Int. batch tank
00:00

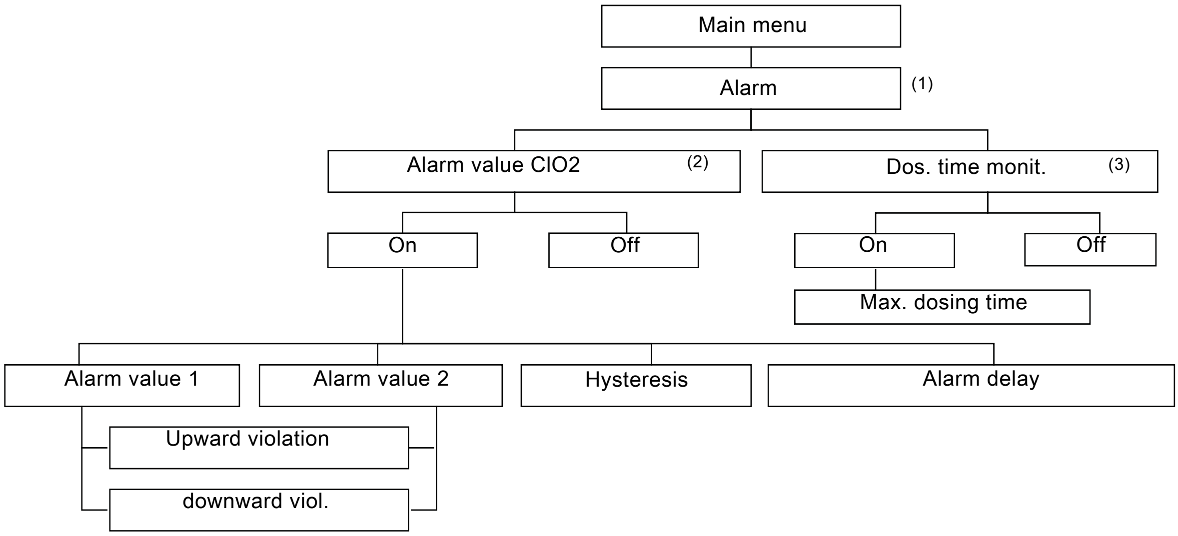
4. Press [Esc].

6.8.10 Testing the display

1. "Main menu > Service > [OK].
2. "Program version" > [OK].

The test function is started. The display goes completely dark, so that each pixel can be checked. In addition, all LEDs are switched on. They light up orange, and the red alarm LED flashes.
After about 5 seconds, the display returns to the "Service > test display" submenu.

6.9 Setting the warning relay and alarm relay



TM03 6937 4506

Fig. 19 Structure of the "alarm" menu

(1) "alarm" is only visible, if "Measurement" or a controller is activated, see "Setup".

(2) "Alarm value ClO2" is only visible, if "Measurement" is activated.

(3) "Dos. time monit." is only visible, if a controller is activated.

English (GB)

6.10 Configuring the warning relay

The warning relay is activated whenever an alarm is signalled, and remains active until the alarm is confirmed
(collective warning relay).

1. "Main menu > Setup" > [OK].
2. "relay" > [OK].
3. Enter the service code, confirm with [OK].
4. "warning relay" > [OK].

fail safe
on (N.C.)
off (N.O.)

The factory setting for the electrically isolated output of the warning relay is "off (N.O.)", normally open.

5. Select "on (N.C.)", confirm with [OK]. The setting is "normally closed". If a power failure occurs, the relay signals a fault (fail safe).

6.10.1 Setting the alarm values

In this menu, you set the upper and lower switching point for the alarm:

1. "Main menu > Alarm" > [OK].

Alarm
Alarm value ClO2
Dos. time monit..

2. "Alarm value ClO2" > [OK].

Alarm value ClO2
Alarm on
Alarm off

The factory setting is "Alarm off".

3. "Alarm on" > [OK].

Alarm value ClO2
Alarm value 1
Alarm value 2
Hysteresis
Alarm delay

Setting the lower switching point for the alarm "Alarm value 1"

1. "Alarm value 1" > [OK].

Alarm value 1
0.15 mg/l

The factory setting is 0.15 mg/l.

2. Factory setting > [Esc], or enter another value > [OK].

eff. direction
Upward violation
downward viol.

The factory setting is "downward viol."

If the alarm is to be triggered by a downward violation:

3. "downward viol.." > [OK]. If an alarm is triggered when the specified value is undershot, the following message appears: "alarm val 2 ClO2 undershot".
   – Rectify the cause of the downward violation.
   – Confirm the alarm message.

If the alarm is to be triggered by an upward violation:

4. "Upward violation" > [OK]. If an alarm is triggered when the specified value is exceeded, the following message appears: alarm val 1 ClO2 exceeded.
   – Rectify the cause of the upward violation.
   – Confirm the alarm message.

Setting the upper switching point for the alarm "Alarm value 2"

1. "Main menu > alarm" > [OK].
2. "Alarm value ClO2" > [OK].
3. "Alarm on" > [OK].

Alarm value ClO2
Alarm value 1
Alarm value 2
Hysteresis
Alarm delay

4. "Alarm value 2" > [OK].

Alarm value 2
0.70 mg/l

The factory setting is 0.70 mg/l.

5. Factory setting > [Esc], or enter another value > [OK].

eff. direction
Upward violation
downward viol.

The factory setting is "downward viol."

If the alarm is to be triggered by an upward violation:

6. Upward violation > [OK]. If an alarm is triggered when the specified value is exceeded, the following message appears: "alarm val 2 ClO2 exceeded".
   – Rectify the cause of the upward violation.
   – Confirm the alarm message.

If the alarm is to be triggered by a downward violation:

7. "downward viol." > [OK]. If an alarm is triggered when the specified value is undershot, the following message appears: "alarm val 2 ClO2 undershot".
   – Rectify the cause of the downward violation.
   – Confirm the alarm message.

6.10.2 Setting the hysteresis

The hysteresis indicates the tolerance of the alarm value (alarm value ± 0.5 x hysteresis). In this menu, you set the hysteresis for the two alarm values:

1. "Main menu > Alarm" > [OK].
2. "Alarm value ClO2" > [OK].
3. "Alarm on" > [OK].
4. "Hysteresis" > [OK].

Hysteresis
0.01 mg/l

The factory setting is 0.01 mg/l.

5. "factory setting" > [Esc], or enter another value > [OK] > [Esc]. The "Hysteresis" function is effective for both alarm value settings.

6.10.3 Setting the alarm delay

1. "Main menu > Alarm" > [OK].
2. "Alarm value ClO2" > [OK].
3. "Alarm on" > [OK].
4. "Alarm delay" > [OK].

Alarm delay
0 sec

The factory setting is 0 seconds.
The setting range is 0-999 seconds.

5. "factory setting" > [Esc], or enter another value > [OK] > [Esc]. The alarm relay is switched on at the end of the specified time.
6. Press [Esc] > [Esc].

6.10.4 Switching on dosing time monitoring

Not for applications with external batch tank.

1. "Main menu > alarm" > [OK].

Alarm
Alarm value ClO2
Dos. time monit.

2. "Dos. time monit." > [OK].

Dos. time monit.
On
Off

The factory setting is "Off".

3. Select "On" to switch on dosing time monitoring, confirm with [OK].

Max. dosing time
600 minutes

The factory setting is 600 minutes.
The setting range is 0-600 minutes.

4. "factory setting" > [Esc], or select another value > [OK] > [Esc]. The alarm is triggered, if the controller sets the predefined maximum dosing flow (Y-OUT = X %) for the dosing pump for longer than the specified time.

6.10.5 Replacing a chemical container

Note

OCD-162-5, -10: Make sure that the chemical containers are positioned under the system.

Warning

Risk of explosion when mixing up chemical containers or suction lances. Possible consequences are personal injury and serious damage to equipment.

Do not mix up chemical containers or suction lances.

Observe the labels on chemical containers, suction lances and pumps:
red = HCl, blue = $NaClO_2$.

Warning

Risk of burns resulting from drips when the suction lance is removed from the chemical container.

Risk of poisoning from chlorine dioxide gas.

Put on protective clothing before starting maintenance work.

Do not allow sodium chlorite and hydrochloric acid to come into contact.

Do not allow drips to fall on the skin, clothing, shoes or ground.

Immediately rinse away any drips on the container or in the collecting tray with water.

Chemical containers should be replaced:

- As soon as possible after the low level signal is displayed.
- Immediately after the empty signal is displayed.

1. Untwist the screw cap from the suction lance in the chemical container.
2. Carefully pull the suction lance out of the container, and immediately insert it into the drip pipe in the collecting tray.
3. If any drips fall onto the container or ground, dilute with water and rinse away immediately.
4. Remove the empty chemical container. Screw the original cover onto the container for storage prior to disposal.
5. Prepare a full chemical container.
6. Unscrew the original cover, and retain it.
7. Insert the suction lance into the full container, and fasten the screw cap again. As soon as the suction lance is inserted, the chlorine dioxide production starts up again. The alarm message is confirmed automatically.

Reset the chemical consumption counter to zero.
For details, see section *6.8.8*.

6.11 Faults with error message

Error message, reaction of Oxiperm Pro	Cause	Remedy
1. Low-level signal HCl or $NaClO_2$: – ClO_2 production proceeds. – Warning relay activated.	a) HCl or $NaClO_2$ container is almost empty.	Change the HCl or $NaClO_2$ container. Oxiperm Pro OCD-162-05, -10: the chemical containers must be positioned under the system.
	b) The float on the suction lance has a wrong orientation.	Turn the float upside down.
2. Empty signal HCl or $NaClO_2$: – ClO_2 production stops, and continues after elimination of the fault. – Alarm relay activated.	HCl or $NaClO_2$ container is empty.	Change the HCl or $NaClO_2$ container. Oxiperm Pro OCD-162-05, -10: the chemical containers must be positioned under the system.
3. "check ClO2 batch": – ClO_2 production proceeds. – Warning relay is activated.	Warning message, undefined contents in the batch tank after power supply failure.	Drain the batch tank manually, and dispose of the content.
4. "level int. batch tank: – ClO_2 production is terminated. – Alarm relay activated.	Too much water is flowing into the batch tank. The ClO_2 solution in the batch tank is too diluted.	Stop the system.
	a) Solenoid valve is leaking.	Check the solenoid valve. Call Service to clean or replace the filter in the solenoid valve.
	b) Faulty float switch in the reaction tank, or too much HCl and/or too much $NaClO_2$ is flowing into the batch tank.	Call Service to replace the float switch in the reaction tank.
5. Max. -Max. level external batch tank: – Alarm relay activated.	Faulty float switch in the external batch tank, or external batch tank is overfull.	Call Service to replace the float switch in the external batch tank.
6. "Timeout H2O supply 1": – ClO_2 production is terminated. – Alarm relay activated.	During the first addition of water (after process start), the level in the reaction tank has increased too slowly. Level K1 was not reached in time.	Call Service.
	a) Filter in solenoid valve clogged, or faulty solenoid valve.	Call Service to replace.
	b) Dilution water supply is not sufficiently open.	Open the dilution water supply further.
	c) Float switch in the reaction tank damaged.	Call Service to replace the float switch in the reaction tank.
7. Timeout pump HCl: – ClO_2 production is terminated. – Alarm relay activated.	During HCl supply, the level in the reaction tank has increased too slowly between levels K1 and K2. Level K2 was not reached in time.	Check the hose from pump to reaction tank for assembly faults. Call Service.
	a) Insufficient performance of the HCl pump – Air in suction hose and/or dosing head. – Pump is not dosing. – Discharge hose is leaking, clogged, porous or bent.	Check the discharge hose. Call Service to replace.
	b) HCl pump is not sucking in – Suction hose is leaking, clogged, porous or bent. – Deposits at the foot valve. – Valve is not installed correctly or clogged. Crystalline deposits in the valves. – Diaphragm is broken (leaking). – Valve tappet is torn out. – HCl container is empty.	• Check the suction hose and suction lance. • Call Service to clean or replace the foot valve. • Call Service to clean the valves. • Call Service to replace the diaphragm. • Check the fill level of the HCl container. • If "empty HCl" is signalled, replace the HCl container.
	c) Flow in the pump not OK.	Vent the system.
	d) Pump is not running at all.	Call Service.
	e) Cable breakage at the controller.	Check the cable from pump to controller. Call Service.
	f) Faulty controller.	Check the controller. Call Service to replace.
	g) Float switch in the reaction tank damaged.	Call Service to replace the float switch in the reaction tank.
8. "Timeout pump NaClO2" – ClO_2 production is terminated. – Alarm relay activated.	During $NaClO_2$ supply, the level in the reaction tank has increased too slowly between levels K2 and K3. Level K3 was not reached in time.	Check the hose from pump to reaction tank for assembly faults. Call Service.
	a) Insufficient performance of the $NaClO_2$ pump. For other reasons, see alarm message Timeout pump HCl.	See alarm message *7. Timeout pump HCl:*.

Error message, reaction of Oxiperm Pro	Cause	Remedy
9. "Timeout H2O supply 2": – ClO_2 production is terminated. – Alarm relay activated.	During the second addition of water, the level in the reaction tank has increased too slowly between levels K3 and K4. Level K4 was not reached in time.	Call Service. See error message *6.*
	a) See alarm message Timeout H2O supply 1.	
10. "Timeout process": – ClO_2 production is terminated. – Alarm relay activated.	After overflow, the level in the reaction tank has dropped back down to K1 too slowly.	
	a) Air bubbles in the overflow pipe.	Vent the system.
	b) Insufficient water supply.	See alarm message Timeout H2O supply 1.
11. "Timeout overflow": – ClO_2 production is terminated. – Alarm relay activated.	During the third addition of water, no overflow from the reaction tank into the batch tank was determined.	
	a) Water supply and solenoid valve.	Call Service.
	b) Air bubbles in the overflow pipe.	Vent the system.
12. "temperature error": – ClO_2 production proceeds.	The temperature at the measuring cell has exceeded the set measuring range.	
	a) Faulty temperature sensor.	Check the temperature sensor. Call Service to replace.
	b) Faulty cable of temperature sensor.	Check the cable of the temperature sensor. Call Service to replace.
	c) Water temperature higher or lower than measuring range.	Check the water temperature.
	d) Temperature measuring range set incorrectly.	Call Service to correct the measuring range.
13. "slope error" – ClO_2 production proceeds.	Plausibility check of calibration data. Calibration fault at calibration level.	Repeat the calibration. Call Service to clean the measuring cell or to replace the electrodes.
14. "fault of electrode/buffer": – ClO_2 production proceeds.	Auto reading of buffer data. Calibration fault at calibration level.	Repeat the calibration. Call Service to clean the measuring cell or to replace the pH electrode.
15. Symmetry error: – ClO_2 production proceeds.	Plausibility check the calibration data for asymmetry potential of pH. Calibration fault at calibration level.	Repeat the calibration. Call Service to clean the measuring cell or to replace the pH electrode.
16. pH buffer difference error: – ClO_2 production proceeds.	Two buffers with a pH difference of less than 1 pH were selected with buffer selection "other". Calibration fault at calibration level.	Check the buffer solutions. Repeat the calibration, and call Service to replace the electrode.
17. Calibration time exceeded: – ClO_2 production proceeds.	Buffer timeout. Fault during pH and ORP calibration. An alarm is activated, if the calibration process does not have a stable measured value, if the time has elapsed. Calibration fault at calibration level.	Check the pH electrode and call Service to replace, if necessary.
18. Offset error: – ClO_2 production proceeds.	Calibration fault at calibration level. Only during ORP calibration.	Repeat the ORP calibration, or call Service to replace.
19. Calibrate ORP electrode: – ClO_2 production proceeds.	The set monitoring time is reached for the next calibration process (calibration interval).	Calibrate or call Service to replace the electrode.
20. "Water sensor fault": – ClO_2 production proceeds. – Combined and setpoint controller stop, and restart after fault elimination.		Stop the controller.
	a) The float of the measuring cell is above the water sensor. The flow is too big.	Reduce the flow with the adjustment spindle of the measuring cell.
	b) The float of the measuring cell float is below the water sensor. The flow is too small.	Increase the flow with the adjustment spindle of the measuring cell.
	c) The sample-water extraction device or the hose to the measuring cell is clogged or leaking.	Check the sample-water extraction device and the hose to the measuring cell.
	d) No sample-water flow in the measuring cell. Filter clogged.	Clean the filter of the measuring cell.
	e) Lack of water at the sample-water extraction device.	Check the flow in the main line at the sample-water extraction device.
	f) Faulty water sensor.	Call Service to replace the water sensor.
	g) Faulty cable from the measuring cell to the controller.	Call Service to replace the cable.
	h) Faulty controller.	Call Service.
	i) Setting in the "Maintenance > water sensor > N.C./N.O." menu does not correspond with terminal connections.	Setting can only be corrected with the super user code.

Error message, reaction of Oxiperm Pro	Cause	Remedy
21. Cleaning motor fault: – ClO_2 production proceeds. – Alarm relay activated. – Combined and setpoint controller stop, and restart after fault elimination.	Cleaning motor monitoring in the measuring cell indicates a fault.	Stop the system.
	a) Faulty cleaning motor.	Check the power supply of the cleaning motor. Call Service to replace the cleaning motor.
	b) No power supply to the cleaning motor. Cable breakage.	Check the cable. Call Service to replace the cable.
	c) Gas bubbles in the measuring cell.	Vent the measuring cell.
22. "dosing time ClO2 exceeded": – ClO_2 production proceeds. – Alarm relay activated. – Controller stops the ClO_2 dosing pump, until the fault is eliminated.	The controller predefines maximum dosing flow for a longer period than the set time.	
	a) After a power supply failure, the solution in the batch tank has been diluted too much after flushing (setpoint and combined controllers).	Continue operation after flushing.
	b) Poor water quality (setpoint and combined controllers).	Measure the water quality and the chlorine dioxide concentration in the main line.
	c) Faulty water flow meter, or wrong setting of water flow meter (proportional and combined controllers).	Call Service to replace.
	d) Faulty measuring cell cable or measuring cell.	Check the measuring cell cable. Call Service to replace.
	e) HCl or $NaClO_2$ container contains only water.	Replace the HCl or $NaClO_2$ container.
	Controller is set incorrectly.	Call Service to check the controller settings.
23. "wire breakage current output 2": – ClO_2 production proceeds. – Alarm relay activated. – Combined and setpoint controller stop, and restart after fault elimination.	The measured chlorine dioxide value can no longer be transmitted.	
	a) Cable breakage at the current output.	Call Service to replace.
	b) Faulty controller.	Call Service to replace the controller.
24. "wire breakage current output 1": – ClO_2 production proceeds. – Alarm relay activated.	Cable breakage at the controller output for the external dosing pump.	
	a) Cable breakage.	Call Service.
	b) Faulty controller.	Call Service to replace the controller.
25. External error: – ClO_2 production is terminated. – Alarm relay activated. – Controller stops the ClO_2 dosing pump, until the fault is eliminated.	An external device, which is possibly connected to a fault input (terminal 51/52), indicates a fault.	
	a) Faulty external device.	Check the external device.
	b) Faulty cable to the external device.	Call Service to check the cable to the external device and to replace, if necessary.
	c) Faulty controller.	Call Service to replace the controller.
26. "Annual maintenance due": – ClO_2 production proceeds.	Maintenance overdue for 0-30 days.	Call Service. The alarm message disappears, if maintenance has been approved.
27. "K5": – ClO_2 production proceeds.	Maintenance overdue for more than 30 days.	Stop the system and call Service.
28. "empty signal int. batch tank": – ClO_2 production proceeds. – Warning relay activated. – Controller stops the ClO_2 dosing pump after 20 seconds, until the fault is eliminated.		
	a) The error message appears in "Int. batch tank" mode (method 1-20 batches). No other process is running.	Check the operation mode. In "Int. batch tank" mode (method 1-20 batches), this is not a fault.
	b) The error message appears, if the dosing pump empties the batch tank before the final chlorine dioxide is available in the reaction tank.	If the message appears in every process, the controller must be reset. In the monitoring menu, check the measured value under "Service > Measurement". In the "Service > Controller ClO2" menu, check the displayed parameter. In the "Service > Water flow meter" menu, check the displayed values.
	c) Faulty water flow meter.	Call Service to check the water flow meter and to replace, if necessary.
	d) Drain cock at the batch tank is open.	Close the drain cock.
	e) Faulty measuring cell (setpoint and combined controllers).	Check the measuring cell. Call Service to replace the measuring cell.

Error message, reaction of Oxiperm Pro	**Cause**	**Remedy**
29. Reaction tank control fault: – ClO_2 production is terminated. – Alarm relay activated.	Plausibility check of the float switch in the reaction tank.	
	a) Faulty float switch.	Call Service.
	b) Faulty controller.	Call Service to replace the controller.
30. Batch control fault: – ClO_2 production is terminated. – Alarm relay activated.	Plausibility check of the float switch in the batch tank.	
	a) Faulty float switch.	Call Service.
	b) Faulty controller.	Call Service to replace the controller.
31. "fault current input": – ClO_2 production proceeds. – Alarm relay activated. – Combined controller and proportional controller stop.	Cable breakage at current input 1 • A water flow meter is connected, and the signal exceeds the full-scale value of 20 mA. • A water flow meter with 4-20 mA is selected, and the signal falls below 3.8 mA. If this fault appears, the controller is stopped (proportional and combined controllers).	Call Service
	a) Faulty water flow meter.	Call Service to check the water flow meter and to replace if necessary.
	b) Faulty current input or controller.	Check the current input and controller. Supply with a defined current between 0 and 20 mA, and compare with display in the "Main menu > Service > Water flow meter" menu. In case of damaged controller, call Service.
	c) Cable breakage between water flow meter and controller.	Call Service to replace the cable.
	d) Water flow meter is connected with 0-20 mA, but the set value is 4-20 mA.	Call Service to correct the software setting.
32. ClO_2 alarm value 1 or 2 exceeded or not reached: – ClO_2 production proceeds. – Alarm relay activated.	The set lower or set upper switching point for the alarm is exceeded or not reached.	Call Service.

English (GB)

6.12 Faults without error message

Fault	Cause	Remedy
The chlorine dioxide dosing pump stops. The DDA pump displays "ERROR".	The isolating valve of the dosing line is closed.	Open the isolating valve. If the Oxiperm Pro operates in 60 Hz mode, check if the multifunction valve is set to 6 bars at the overflow side. See also the installation and operating instructions of the multifunction valve.
Overdosing of chlorine dioxide dilution due to a free discharge.	The chlorine dioxide dosing pump has a free discharge into a container. Even if the pump stops, the chlorine dioxide solution continues running into the container due to a siphon effect. The consequence is overdosing. The conditions are: • disconnected injection unit • dosing pump without multi-function valve.	Connect the multifunction valve at the pump. This will prevent uncontrolled flow of the dosing liquid through the dosing lines.
Perceptible smell of chlorine dioxide	The activated carbon filter/adsorption filter is saturated.	Exchange the activated carbon filter/adsorption filter.

6.13 Calibration

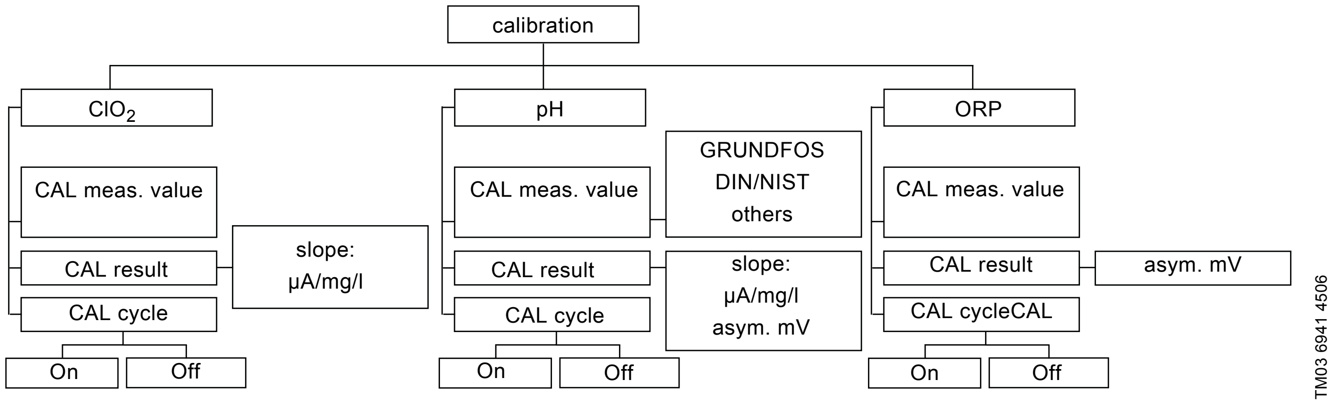


Fig. 20 Structure of the "calibration" menu

Note

Perform calibration only with constant measured values from the measuring cell. For details, see installation and operating instructions for the measuring cell.

Prior to calibration, check the measured value of the electrode of the measuring cell.

6.13.1 Calibrating the chlorine dioxide measured value

To calibrate the chlorine dioxide measured value, a reference measurement must be taken. This can be made with the Grundfos DIT photometer and the usual reagents.

The determined reference value is entered by correcting the current measured value in the display.

The controller reads in the new measured value. It assigns the current signal [μA] from calibration, coming in at the current input of the measuring cell, to the new measured value.

Check, if the measured value of the electrode is constant:

1. "Service > Measurement > ClO2 > Measured value".
   – Current chlorine dioxide concentration at the measuring cell
   – Current signal of the measuring cell
   – Measuring range.

measurement ClO2
0.21 mg/l
5.800 μA
0.0 - 0.5 mg/l

If the measured value remains constant:

2. Determine the chlorine dioxide value by means of a reference measurement, and note it down.
3. Press [Cal].

calibration
chlorine dioxide
pH/ORP

4. "chlorine dioxide" > [OK].

chlorine dioxide
CAL meas. value
CAL result
CAL cycle

5. "CAL meas. value" > [OK].

CAL meas. value
0.05 mg/l
I-cell: 5.2 μA

6. Set the determined reference value in mg/l with [Up] or [Down], confirm with [OK]. The controller assigns the reference value to the current signal.
7. "CAL cycle" > [OK].

CAL cycle
slope:
22.0 μA/ppm

The slope is the straight line between the measured value and the zero point. The unit is μA per ppm (ppm = parts per million = mg/l in water).

The slope can be presented graphically:

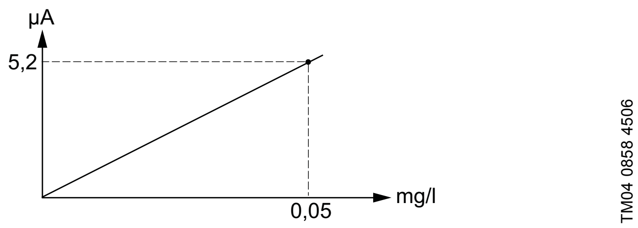


Fig. 21 Reference value measured with e.g. the DIT photometer

From now on, the controller uses this value as the basis for calculations. Chlorine dioxide calibration is completed.

Reading the slope in the calibration logbook

1. "Service > Measurement " > [OK].
2. "ClO2" > [OK].
3. "CalData-LogBook" > [OK]. Entry number 1 is the latest entry, entry number 2 is the previous one, etc.

CalData-LogBook
number...............................1
Date....................2007-07-3 1
Time............................12:3 4
slope:...................... 22.0 μA

Enabling and disabling the display of the chlorine dioxide calibration interval

1. "calibration > ClO2 > CAL cycle" > [OK].
2. (Interval) Off > [OK] or (Interval) On > [OK].

6.13.2 Performing (two-point) pH calibration

The electrode sends the voltage [mV] that corresponds to the pH value to the controller. Two different buffer solutions can be used to calibrate the pH measured value.

1. Prepare two glass jars with the buffer solutions.
2. Prepare an empty 10-litre plastic bucket.
3. Measure the temperature of the buffer solution.
4. Press [Cal].

calibration
chlorine dioxide
pH/ORP

5. pH > [OK].

pH
CAL meas. value
CAL result
CAL cycle

6. "CAL meas. value" > [OK].

CAL meas. value
GRUNDFOS
DIN/NIST
others

7. Select one of the three buffer types with [Up] or [Down].

Buffer type	Buffer values
GRUNDFOS	4.01, 7.00, 9.18
DIN/NIST	4.01, 6.86, 9.18
others	The lower and upper buffer values can be freely adjusted (difference of at least 1 pH) within the set pH measuring range.

8. GRUNDFOS > [OK].

buffer temp.
25 °C

9. Set the measured temperature of the buffer solution with [Up] or [Down], confirm with [OK].
10. Shut off the water supply of the measuring cell.

11. Unscrew the pH electrode from the measuring cell. Use the bucket to catch any water that flows out.
12. Immerse the pH electrode in the glass jar containing the second buffer solution (for example 4.01 pH).

buffer value
4.01 pH
7.00 pH
9.18 pH

13. Select the buffer value of the buffer solution in which the electrode is immersed (for example 4.01 pH). The voltage [mV] at the electrode in the second buffer solution (for example 4.01 pH) is measured, and assigned to the pH value.
14. Remove the pH electrode from the buffer solution, and rinse with water.
15. Immerse the pH electrode in the glass jar containing the second buffer solution (for example 7.00 pH).

buffer value
4.01 pH
7.00 pH
9.18 pH

16. Select the buffer value of the buffer solution in which the electrode is immersed (for example 7.00 pH). The voltage [mV] at the electrode in the second buffer solution (for example 7.00 pH) is measured, and assigned to the pH value.

The slope is the straight line between both measured values. The unit is mV/pH.

The slope can be presented graphically:

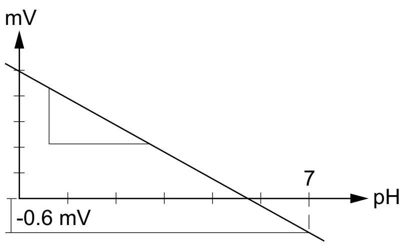


Fig. 22 Performing the two-point pH calibration

17. CAL result > [OK]. The result is displayed as the slope of the straight line, and the asymmetry (the asymmetry is the deviation from the zero point at pH 7). In this example, one pH unit corresponds to -57.88 mV.

CAL result
slope: -57.88 mV/pH
asym. -0.6 mV

18. Remove the pH electrode from the buffer solution, and rinse with water.
19. Screw the pH electrode back into the measuring cell. The current pH value of the water in the main line is updated in the display level.
20. Open the water supply of the measuring cell.
21. Dispose of the buffer solution. Do not pour it back into the bottle.
22. Pour the contents of the bucket down the drain. pH calibration is completed.

For alternative method, see the operating and installation instructions for the respective measuring cell.

Enabling and disabling the display of the pH calibration interval

1. calibration > pH > CAL cycle > [OK].
2. (Interval) Off > [OK] or (Interval) On > [OK].

6.13.3 Performing ORP calibration

The electrode sends the voltage [mV] that corresponds to the ORP value to the controller. It indicates the voltage of all ions in the water (summation parameter).

1. Prepare a glass jar with ORP buffer solution with a known mV value.
2. Prepare an empty 10-litre plastic bucket.
3. Press [Cal].

calibration
chlorine dioxide
ORP

4. ORP > [OK].

ORP
CAL meas. value
CAL result
CAL cycle

5. "Measured value" > [OK].

CAL meas. value
225 mV

6. Turn off the water supply of the measuring cell.
7. Unscrew the ORP electrode from the measuring cell. Use the bucket to catch any water that flows out.
8. Immerse the ORP electrode in the glass jar containing the ORP buffer solution.
9. In the display, set the mV value of the ORP buffer solution, confirm with [OK]. The mV value of the ORP buffer solution is measured.
10. "CAL result" > [OK]. The "ORP offset value is displayed as the result (for example 2 mV). This is the deviation between the entered mV value and the measured mV value of the buffer solution. The controller corrects the measured ORP value in the water of the main line by the offset value.

CAL result
ORP offset
-2 mV

11. Remove the ORP electrode from the buffer solution, and rinse with water.
12. Screw the ORP electrode back into the measuring cell.
13. Open the water supply of the measuring cell.
14. Dispose of the buffer solution. Do not pour it back into the bottle.
15. Pour the contents of the bucket down the drain. ORP calibration is completed.

For alternative method, see the operating and installation instructions for the respective measuring cell.

Enabling and disabling the display of the ORP calibration interval

1. calibration > ORP > CAL cycle > [OK].
2. (Interval) Off > [OK] or (Interval) On > [OK].

6.13.4 Calibration faults

1. If, for example, 4.0 pH is selected in the display, but the electrode is immersed in a 7.00 buffer solution, the following error message is displayed: "wrong buffer"
   – Press [Esc] to terminate calibration, and repeat the process correctly.
2. If the slope or asymmetry is outside the norm, the following error message is displayed: "slope error", "error asym. pot.". This is caused by an old electrode or buffer solution. Check the expiry date.
   – Press [Esc] to terminate calibration, replace the electrode, and repeat calibration.
3. If the electrode does not send a stable measuring signal to the controller within 120 seconds, the following error message is displayed: "calibration time exceeded". This is caused by an old electrode.
   – Press [Esc] to terminate calibration, replace the electrode, and repeat calibration.

6.14 Stopping chlorine dioxide dosing

The controller can be switched off in the "Setup > Controller ClO2" menu. This is not recommended, because all the controller parameters will have to be reset, when it is restarted.

Switching off the controller in manual operation

1. Press [Man].
   – If the controller is switched off in "Setup", the message "Check settings!" is displayed. You cannot access manual operation.
   – If the controller is switched on in "Setup", the "Man" LED lights up, and "manual operation" is ready for use.

manual operation
Controller ClO2
dosing flow

2. "Controller ClO2" > [OK].

Controller ClO2
On
Off

The factory setting is "On".

3. Off > [OK]. The display automatically reverts to "manual operation". The controller is switched off, and the dosing pump shuts off.

Stopping the controller via the higher-level control system

The controller and the dosing pump can be stopped via an external device, for example a higher-level control system.

Switching off the system at the main switch

The Oxiperm Pro OCD-162 system can be switched off at the main switch. This terminates the production and dosing process. For restart, see section *6.4.4*.

6.15 Switching off the system

1. See section *6.3.1 Terminating chlorine dioxide production*.
2. See section *6.5 Flushing*. The dosing pump is switched off automatically, as soon as the batch tank is empty.
3. Switch off the Oxiperm Pro at the main switch.
4. Close the dilution water supply.

For restart, see section *6.4 Continuing operation after an interruption*.

7. Maintenance

The Oxiperm Pro OCD-162 disinfection system must be serviced once a year. The current and next maintenance date are displayed in the operating software. The first maintenance date is one year after commissioning. Display the maintenance date:

1. Main menu > Service" > [OK].
2. "Process" > [OK].
3. "Production ClO2" > [OK].
4. "Maintenance" > [OK] > [Esc].

For further information, see the separate Service Instructions.

7.1 Cleaning

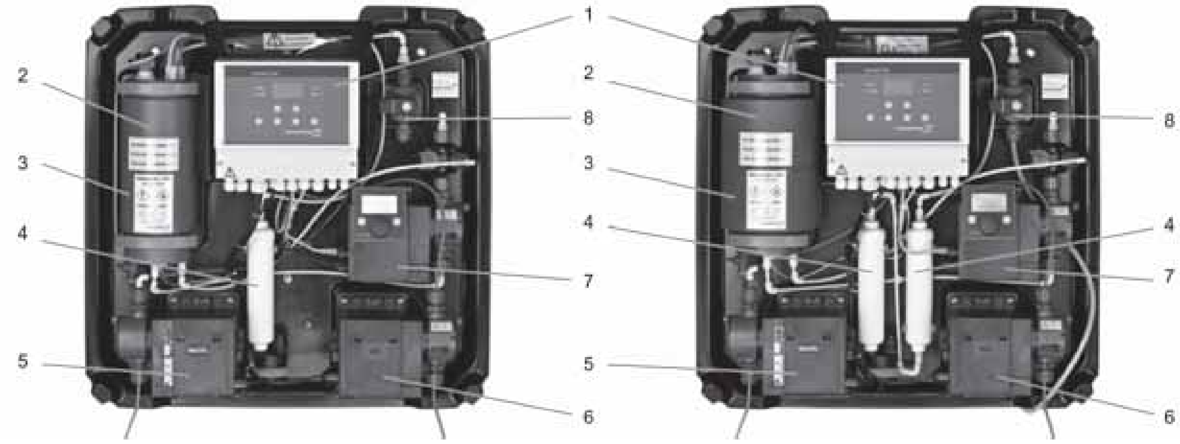
If necessary, clean all pump surfaces with a dry and clean cloth.

8. Accessories list

Prior to installation, the operator must purchase the following accessories according to the product numbers in the Oxiperm Pro OCD-162 data booklet and the technical data.

Accessories to be purchased	
Y = Available from Grundfos N = Not available from Grundfos	
1. Container with sodium chlorite (diluted concentration of 7.5 % by weight in accordance with EN 938) 2. Container with hydrochloric acid (diluted concentration of 9.0 % by weight in accordance with EN 939)	N
3. Two collecting trays for the chemical containers	Y
4. Inductive or ultrasonic flow meter, if necessary	Y
5. Flow meter connection cable, if necessary	Y
For dilution water line (if no mixing module with dilution water connection has been ordered):	
6. Tapping sleeve	Y
7. Extraction device	Y
8. Hose connection for dilution water hose	Y
9. Sample-water filter (in case of insufficient water quality)	Y
For main water line:	
10. Tapping sleeve for injection unit	Y
11. Two tapping sleeves for extension module, if necessary	Y
12. Tapping sleeve for sample-water extraction	Y
Oxiperm Pro hoses:	
13. Hose between dilution water extraction device and solenoid valve	Y
14. Dosing line between dosing pump and injection unit	Y
Measuring cell hoses:	
15. Hose between measuring cell and sample-water extraction device	Y
16. Hose between measuring cell and drain	Y
For mixing module, if installed:	
17. Hose between mixing module and main water line and hose back to mixing module	Y
18. Dosing line between dosing pump and injection unit in mixing module	Y
Or for measuring module, if installed:	Y
19. Hose between measuring module and main water line and hose back to measuring module	Y
20. Protective pipe for dosing hose	N
21. Main switch	N
Cables:	
22. Oxiperm Pro power supply cable	N
23. Power cable for measuring module or mixing module, if necessary	N
24. Protective clothing	Y
25. Two 10-litre plastic buckets	N
26. 100 g sodium thiosulphate (20 gram per flushing process)	N

9. Photos



TM03 6959 4506

Fig. 23 OCD-162-5, -10

Pos.	Component
1	Controller with display and control elements
2	Reaction tank
3	Batch tank
4	Adsorption filter
5	Dosing pump for sodium chlorite ($NaClO_2$)
6	Dosing pump for hydrochloric acid (HCl)
7	Dosing pump for chlorine dioxide (ClO_2)
8	Solenoid valve for dilution water

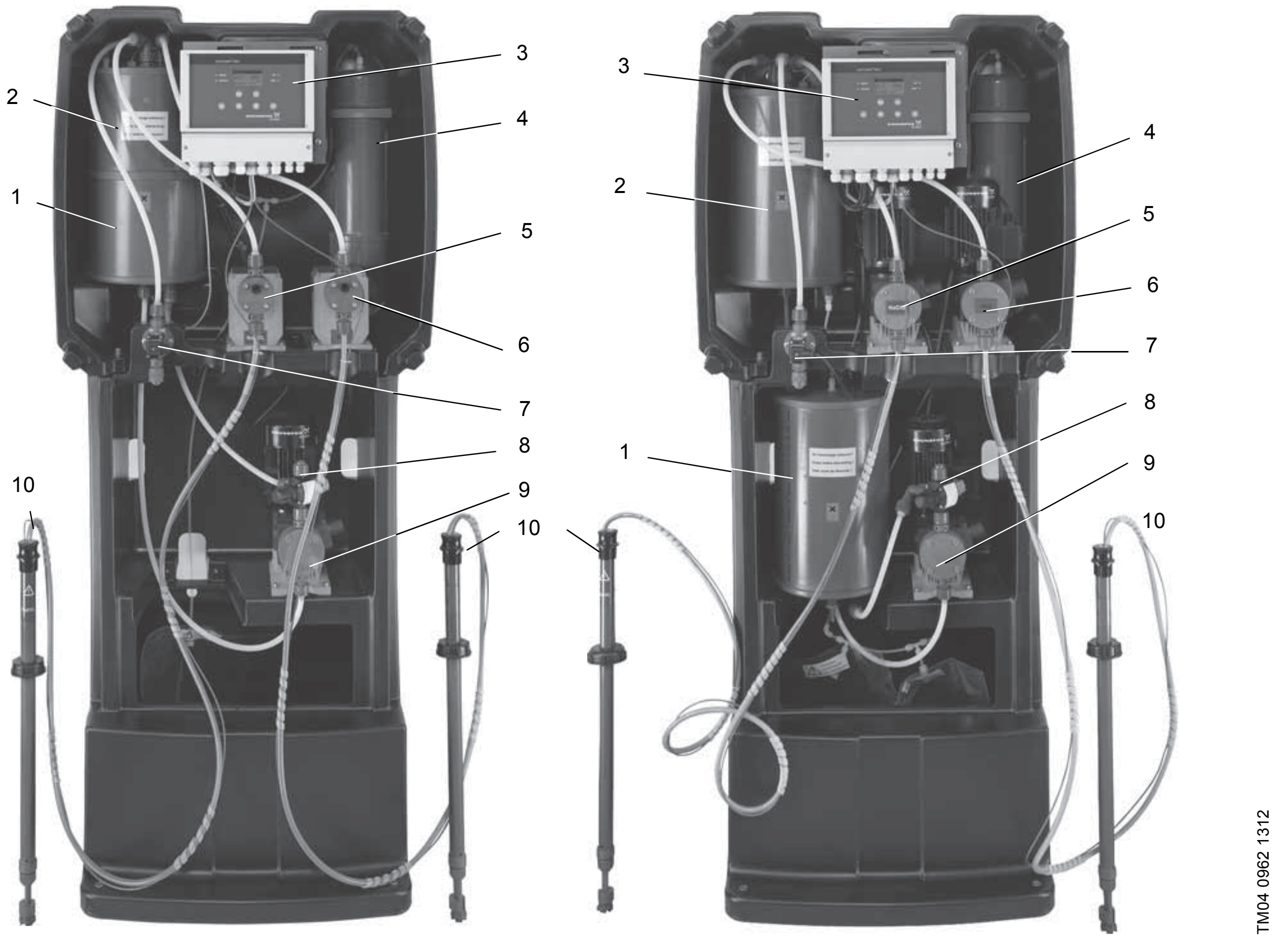


Fig. 24 OCD-162-30, -60

Pos.	Component
1	Batch tank
2	Reaction tank
3	Controller with display and control elements
4	Adsorption filter
5	Dosing pump for sodium chlorite ($NaClO_2$)
6	Dosing pump for hydrochloric acid (HCl)
7	Solenoid valve for dilution water
8	Multi-function valve
9	Dosing pump for chlorine dioxide (ClO_2)
10	Suction lance

10. Disposal

Note

The Oxiperm Pro OCD-162 disinfection system and its associated parts must be disposed of in an environmentally friendly way.

The system may only be dismantled by authorised personnel trained by Grundfos.

The operating company is responsible for ensuring an environmentally friendly disposal.

Before dismantling, the Oxiperm Pro system must be completely rinsed with water in order to remove the chemicals from the reaction tank, the hoses and the pumps. The dosing line must be placed outdoors to let residual chlorine dioxide escape.

For environmentally friendly disposal, the operating company should hand over the Oxiperm Pro system or its parts to a private disposal service. If there is none in your region, send the system to the nearest Grundfos company.

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BE > THINK > INNOVATE >

Being responsible is our foundation
Thinking ahead makes it possible
Innovation is the essence

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